PROPOSED CONSTRUCTION OF KIRIAINI AHP IN MATHIOYA CONSTITUENCY, MURANG'A COUNTY WITH ASSOCIATED AMENITIES AND INFRASTRUCTURE

MURANGA COUNTY

TENDER NO. MLPWHUD/SDHUD/ AHP /303/2024-2025

GEOTECHNICAL INVESTIGATION REPORT

JUNE 2026

MINISTRY OF LANDS, PUBLIC WORKS, HOUSING AND URBAN DEVELOPMENT,
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Table of Contents

1. INTRODUCTION 5
1.1 General Site Location and Description 5
1.1.1 Topographical Condition 6
1.1.2 Surface Water 7
1.1.3 Drainage 7
1.1.4 Climate 7
1.1.4.1 Temperature 7
1.1.4.2 Rainfall 8
1.1.4.3 Humidity 8
1.1.4.4 UV Radiation 8
1.1.4.5 Wind 9
1.1.4.6 Clouds 9
1.1.4.7 Sunshine 9
1.2 Objective and scope of the investigations 10
1.3 Exploratory Borehole locations (Setting Out) 10
2. EXISTING INFORMATION ON GENERAL GEOLOGY OF THE REGION 12
2.1 Geology of Murang'a County 12
2.2 Climate 15
2.3 Temperatures 16
2.4 Precipitation 16
3. FIELD WORKS, TEST MATERIAL SAMPLING AND SAFETY PRECAUTIONS 17
3.1 Drilling of Exploratory Boreholes (Methodology) 17
3.2 Equipment 17
3.3 Standard Penetration Test 17
3.4 Dilatancy Correction Formula 18
3.5 Conversion of SPT to Allowable bearing Capacities 18
3.5 Material Sampling (Soil Sampling) 20
3.6 Subsurface Conditions and Stratigraphy (Specific Geology) 21

3.7 Measurement of Ground Water levels in boreholes 22
3.8 Limitations of the Investigation Method 22
4. LABORATORY MATERIAL TESTING 23
4.1 Determination of Particle Size Distribution (Sieve Analysis) 24
4.2 Determination of Atterberg limits 24
4.3 Determination of Maximum Dry Density/Optimum Moisture Content. 25
4.4 Determination of Direct shear test 26
5. CALCULATION/ESTIMATION OF BEARING CAPACITIES IN SOIL FORMATIONS 28
6. DISCUSSIONS AND CONCLUSIONS 31
7. RECOMMENDATIONS 32
7.1 Slab-On-Grade Drainage 32
7.2 Foundation type and allowable bearing pressure 32
7.3 Earth work fill and earth work cut 34
8. LIMITATIONS AND RISKS 35
9. REFERENCES 37
Disclaimer 38
10. Appendix I (LABORATORY AND INSITU TEST REPORTS) 39
10.1 Determination of Particle Size Distribution 40
10.2 Determination of Atterberg limits 41
10.3 Dry Density Moisture Content relationships 42
10.4 Direct Shear Box Test 43
11. APPENDIX II (BOREHOLE LOGS) 44
12. APPENDIX III (CORE BOX AND SITE PHOTOGRAPHS) 45

List of Figures

Figure 1: Satellite images showing the general location of the study area 6
Figure 2: Geological Map of Murang'a County 14
Figure 3: Pysiography of Murang'a County 15

List of Tables

Table 1: Survey point coordinates Elevations and depths 11
Table 2: SPT N Values and Bearing Capacity Estimation 19
Table 3: Test Procedures and Standards 23
Table 4: Particle size Distribution 24
Table 5: Atterberg limits 25
Table 6: Dry density and Moisture content relationship 26
Table 7: Direct shear strength 27
Table 8: Bearing capacities computed from direct shear test parameters (Overburden of 1.5m) 29
Table 9: Bearing capacities computed from direct shear test parameters (Overburden of 2.0m) 29
Table 10: Terzhagi's constants 29

1. INTRODUCTION

STATE DEPARTMENT OF HOUSING AND URBAN DEVELOPMENT retained **CRANSFIELD MATERIAL TESTING CENTRE** to conduct a Geotechnical Site Investigation to obtain information on the general ground conditions and stratigraphy at the Proposed Construction of Kiriaini AHP in Mathioya Constituency, Murang'a County with Associated Amenities and Infrastructure.

The works comprised drilling Two (2No.) boreholes with the aim of identifying the distribution of subsurface geotechnical soils/rocks and engineering geological properties at the proposed project area so as to provide reliable geotechnical data for engineering design.

The site investigations fieldwork was undertaken from 4TH to 9TH June 2026.

Recovered samples were safely transported and submitted to our laboratory in Nairobi for analysis.

This document therefore reports the findings from the borehole logs and the laboratory analysis, and has been issued upon completion of fieldwork at the project site, and laboratory analysis.

All the fieldwork was carried out in accordance with BS 5930: 2015 (code of practice for site investigations). Laboratory tests were done as stipulated in the relevant standards and procedures; BS 1377/ASTM & ISRM

1.1 General Site Location and Description

The project area is located on a public land off Kangema – Kiriaini – Othaya Rd in Kiriaini, Murang'a County. The economy of Kiriaini is strongly supported by its strategic infrastructure, which facilitates high-volume agricultural productivity and cross-county trade. Positioned along the critical C70 bitumen corridor, Kiriaini serves as a vital transit hub, enabling the efficient transport of major cash crops—predominantly tea, coffee, and fresh horticulture—from the surrounding deep-ridge farms directly to major processing centers like the Kiru and Gatunguru tea factories. This reliable road network significantly lowers transit times and overheads for local

farmers, while an expanded electricity grid fuels essential agro-processing operations, cold storage initiatives, and a bustling commercial center. Robust water supply projects and accessible local social amenities further sustain both the intensive farming activities and the surrounding rural workforce, solidifying infrastructure as the primary driver of Kiriaini's thriving, agriculture-led economy.

The site co-ordinates are: - (*LATITUDE: 0°36'32.90"S, LONGITUDE: 36°56'10.50"E)*

Below in figure No.1 is a satellite image showing the general location of the investigation area.

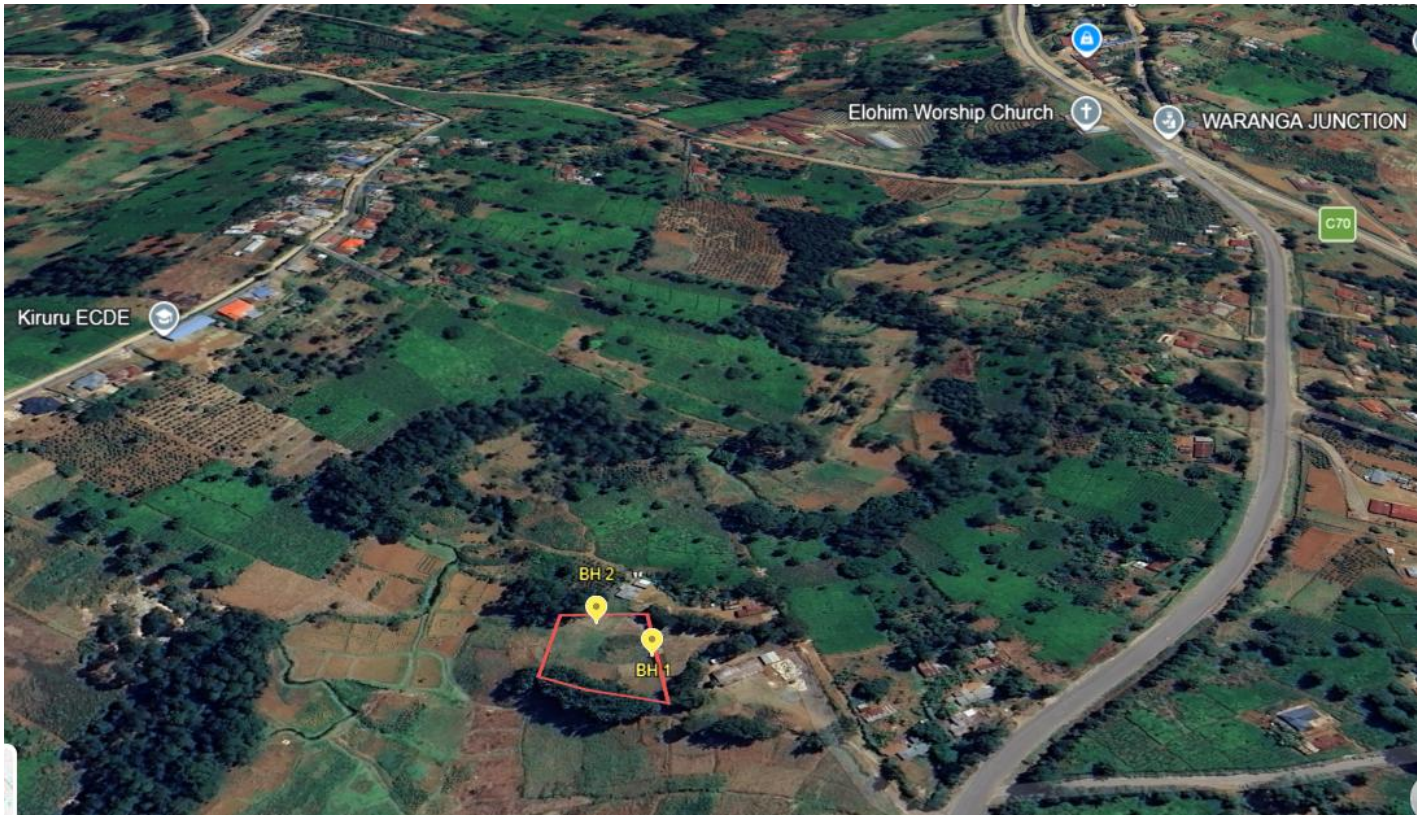


Figure 1: Satellite images showing the general location of the study area

1.1.1 Topographical Condition

The topography of Kiriaini is characterized by gently undulating terrain with a mix of low hills and broad valleys. The area generally slopes eastwards toward the lower plains, with elevations decreasing from the central highlands of Murang'a. It is well-drained by several seasonal and

permanent streams that support agriculture, and the landscape is largely suitable for farming, particularly coffee and tea cultivation, due to its moderate slopes and fertile soils.

The project is on a gentle to medium sloppy terrain towards the Northern side of the site and features some trees and plants.

1.1.2 Surface Water

No surface water was encountered at the proposed site during, and throughout the time of the investigations.
This is a clear indication that the site is well drained.

1.1.3 Drainage

No storm water drainage infrastructure has been developed. We recommend the design team to consider draining the storm water to recharge pits or develop storm water drainage system as budget may allow.

The project area has not been connected to a sewer line therefore for purposes of project planning, the client should budget for an on-site water treatment system.

Rain water collection from any pitched roof can also be harvested in an underground water tank. No ponding of water was observed on site.

1.1.4 Climate

1.1.4.1 Temperature

Kiriaini experiences a tropical savanna climate, characterized by warm temperatures throughout the year with distinct wet and dry seasons. During the wet season, which typically falls between March and May, the area receives abundant rainfall, leading to lush vegetation and higher humidity levels. Temperatures during this period remain relatively warm, with average highs around 27°C (81°F) and lows around 18°C (64°F). The dry season, spanning from June to February, is marked by lower rainfall and drier conditions. Temperatures during this period can be quite hot,

with average highs reaching 30°C (86°F) and lows around 15°C (59°F). However, the dry air and lower humidity often make the heat feel more tolerable compared to the humid wet season.

1.1.4.2 Rainfall

Kiriaini experiences a bimodal rainfall pattern, with two distinct rainy seasons. The first rainy season occurs between March and May, during which the area receives significant rainfall. The second rainy season takes place between October and December, although it is generally less intense than the first. On average, Kiriaini receives approximately 1,200 millimeters (47 inches) of rainfall annually. The highest rainfall amounts are typically recorded in April, with an average of around 230 millimeters (9 inches). November is the second wettest month, receiving an average of about 160 millimeters (6 inches) of rainfall. The driest months are January and February, with minimal rainfall during this period.

1.1.4.3 Humidity

Kiriaini in Murang'a County experiences varying humidity levels throughout the year, influenced by its tropical savanna climate and seasonal rainfall patterns. During the wet seasons, particularly March to May and October to December, humidity levels tend to be higher due to increased rainfall and atmospheric moisture. Relative humidity can reach levels above 70% during these periods, making the air feel more humid and potentially uncomfortable.

In contrast, during the dry season (June to February), humidity levels decrease significantly as rainfall diminishes. Relative humidity can drop to as low as 30% during the peak of the dry season.

1.1.4.4 UV Radiation

Kiriaini in Murang'a County, Kenya, experiences high levels of UV radiation, particularly during the dry season (June to February). This is due to its location near the equator, high altitude, and clear skies during this period. The UV index, a measure of the strength of UV radiation, often reaches levels of 8 or higher, which is considered very high. From a design perspective, these conditions necessitate the incorporation of effective solar control measures such as shading devices, deep roof overhangs, and the use of UV-resistant materials. Proper orientation of buildings, limited direct sun penetration through windows, and landscaping elements like tree

cover can further help reduce solar heat gain and enhance thermal comfort.

1.1.4.5 Wind

Kiriaini in Murang'a County generally experiences moderate wind speeds throughout the year. The prevailing wind direction is easterly, influenced by the prevailing trade winds. Average wind speeds typically range from 5 to 15 kilometers per hour (3 to 9 miles per hour). However, during the dry season (June to February), stronger winds can occur, particularly in the afternoon and evening. These winds can sometimes reach speeds of 20 to 30 kilometers per hour (12 to 19 miles per hour). When designing for wind loading in Kiriaini , engineers should refer to site-specific meteorological data and applicable structural design codes to accurately determine wind speeds and pressures. Due to the area's exposure and elevation, it is important to incorporate wind-resistant design considerations such as adequate structural bracing, secure roof anchorage, and proper connections between structural elements to enhance overall building stability and safety.

1.1.4.6 Clouds

Kiriaini in Murang'a County experiences varying cloud cover throughout the year, influenced by its tropical savanna climate and seasonal rainfall patterns. During the wet seasons, particularly March to May and October to December, cloud cover tends to be higher, with frequent overcast skies and occasional rainfall. This increased cloud cover helps to moderate temperatures and reduce the intensity of sunlight.

In contrast, during the dry season (June to February), cloud cover is significantly less, leading to more sunshine and higher temperatures. Clear skies are common during this period, allowing for ample sunlight and reducing the moderating effect of clouds. However, it's important to note that occasional cloud cover can still occur, especially during the transition periods between seasons.

1.1.4.7 Sunshine

Kiriaini in Murang'a County enjoys a significant amount of sunshine throughout the year, especially during the dry season (June to February). This is due to its location near the equator, high altitude, and clear skies during this period. On average, Kiriaini receives around 7-8 hours of

sunshine per day during the dry season. However, during the wet seasons (March to May and October to December), cloud cover can reduce the amount of sunshine, with daily averages ranging from 4 to 6 hours.

1.2 Objective and scope of the investigations

The following were the objectives of the geotechnical survey:

a) Drill 2No. exploratory boreholes with depths to a maximum depth of 25 meters to realize a total meter run of 50.0 meters below the original ground level as shown below in table No. 1
b) Determine the subsurface stratigraphy to the full depths of the boreholes by logging and monitoring of ground water levels.
c) Sampling of undisturbed and/or disturbed test materials; soil/rocks
d) Conduct engineering material analysis on the recovered test samples
e) Provide lithological descriptions of the stratigraphic units encountered in the boreholes in form of logs.
f) Evaluate the engineering characteristics of the subsurface materials through in-situ and laboratory testing of the representative samples recovered from the zone of influence of foundation loadings.
g) Prepare an interpretative geotechnical report for the site, to provide clear geotechnical recommendations for safe and economical design of the proposed development.

This exercise, in entirety was conducted with the aim of evaluating the subsurface ground conditions by finding out their engineering properties, in terms of classification and strength characteristics in order to aid the design engineer on technical decision making regarding the probable foundation type, founding depth and size.

1.3 Exploratory Borehole locations (Setting Out)

The drilling points of the boreholes were identified by the client's structural engineer. A hand-held GPS device was then used to locate the actual drilling points on ground. The 2 exploratory boreholes were set out as summarized below in Table No.1.

Table 1: Survey point coordinates Elevations and depths

Borehole ID	DEPTH (M)	Placed Coordinates		Elevations (M)
		Latitudes	Longitudes	
BH 01	25.0	0°36'32.90"S	36°56'10.50"E	1883.4
BH 02	25.0	0°36'32.00"S	36°56'10.70"E	1880.8

2. EXISTING INFORMATION ON GENERAL GEOLOGY OF THE REGION

2.1 Geology of Murang'a County

The geology of 36°56'10.70"E and that of Fort Hall in Murang'a County are similar and spans several geological periods which can be broadly divided into four primary geological units based on the age and type of rocks present, according to the Geological Survey of Kenya (1966):

1. **Recent:** Soils and alluvium
2. **Pleistocene:** Sandstones, Thiba basalts
3. **Tertiary:** Tumu Tumu trachytes, Mount Kenya phonolites, Laikipian basalts, Trachytic tuffs, Simbara Series (porphyritic olivinebasalts and basaltic agglomerates) and Kapiti phonolite
4. **Precambrian:** Metamorphosed and granitized rocks of the Basement System

Basement System rocks are mainly exposed in the south-eastern quarter of the area and also appear as inliers protruding through the volcanic rocks near Fort Hall. They were formed by recrystallization of original mixed sedimentary sandstones and shales into which lenses and sills of basic igneous rocks were intruded.

The long period of time intervening between the metamorphism and recrystallization of the Basement and the first extrusion of Tertiary lavas was one of repeated uplift and erosion attended by development of the peneplained erosion surfaces already described.

At the beginning of the Miocene the Fort Hall area must have been an undulating plain studded with minor hills such as Mutheru and Molinduku with a higher group of hills rising some 1,200 feet above plain-level in the present area of the Ithanga and Kakuzi hills. Before extrusion of the earliest Tertiary lavas the plain was uplifted and a considerable valley was incised along the present site of the Maragua gap to the south of Fort Hall, at the same time the southern margin of the Muhito inlier stood as a scarp overlooking the valley.

The first lava flood of Kapiti phonolite erupting from the eastern flank of the Rift Valley was evidently halted by the barrier of the Ithanga and Kakuzi hills and continuing eastward flow was diverted through the Maragua gap, so that in this area the sub-Miocene surface was not covered by Kapiti phonolite. The bulk of the succeeding Simbara basalts and agglomerates are also

restricted to the west of the Kakuzi hills and the Muhito inlier. The effectiveness of the latter as a barrier is illustrated by outliers of Simbara agglomerate near Fort Hall and the overlap of younger porphyritic basalts on older Simbara agglomerate near the Gura river. The passage of the Simbara basalts prob˜bly caused final filling of the Maragua gap.

The end-Tertiary erosion surface was well developed before the deposition of trachytic tuffs overlying the Simbara volcanics west of the Kakuzi-Muhito barrier, since similar tuffs rest directly on Basement System at a much lower level to the east of this barrier. Laikipian basalt then flowed into the area from the north-west, entering the area around the northern end of the Muhito inlier, and was followed by the first eruption of lavas from Mount Kenya. The first flows were of phonolite which rest directly on the Basement System, and were followed by trachytes entering the area from the north-west.

A long period of erosion following the emplacement of the trachytes led to maturation of the end-Tertiary surface over an extensive area between the Muhito inlier and the Mount Kenya phonolite north of the Tana River. Succeeding lava flows from Mount Kenya (the Thiba basalts) flooded the whole of this part of the end-Tertiary plain and filled many of the river valleys excavated in older volcanics. Their deposition on the mature end-Tertiary surface dates the Thiba basalts as Pleistocene. Damming of drainage from the Aberdares by the Thiba basalts led to accumulation of silts and sandstones covering an area of several square miles to the west of Kiambicho.

Tertiary to Recent tectonic activity in the area was slight; faulting is mainly confined to the oldest Tertiary lavas (the Kapiti phonolite), and in recent times geological processes have been limited to soil formation, excavation of river valleys and the re-excavation of Basement System inliers.

Figure 2 bellow shows the map of the study area.

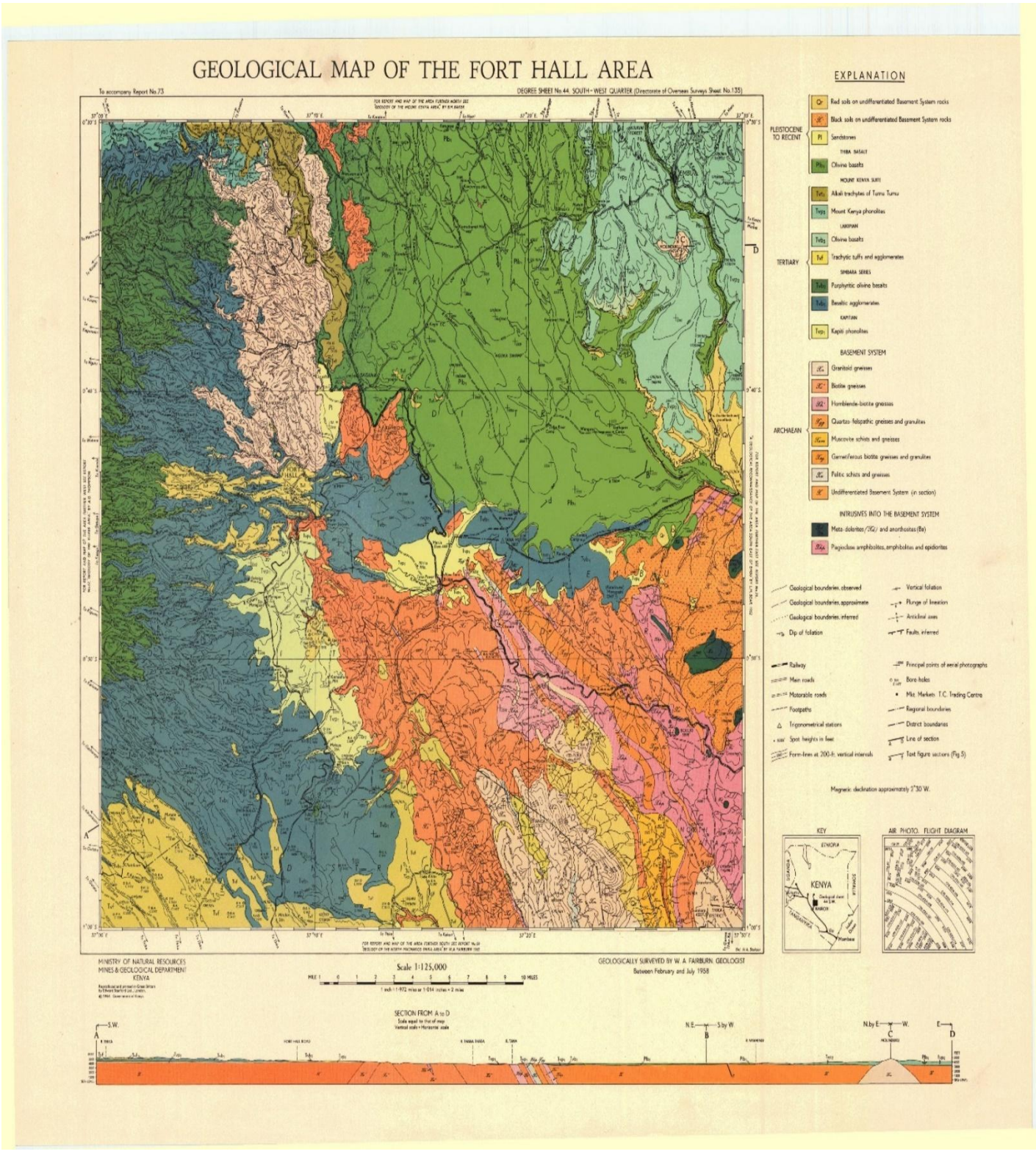


Figure 2: Geological Map of Murang'a County

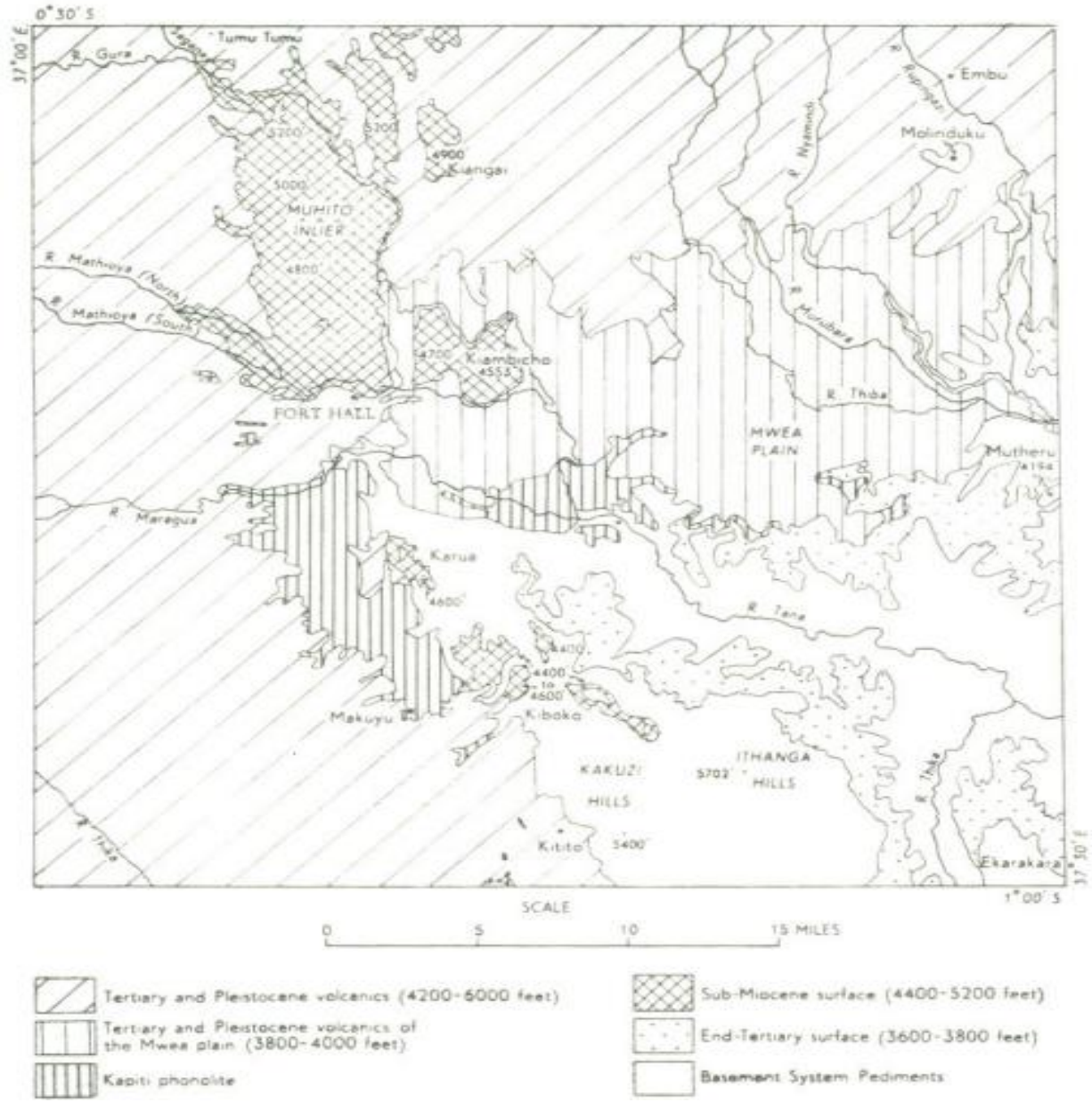


Figure 3: Pysiography of Murang'a County

2.2 Climate

36°56'10.70"E in Murang'a County, Kenya, experiences a tropical savanna climate, characterized by distinct wet and dry seasons. The average annual temperature ranges from 15°C (59°F) to 30°C (86°F). The wet seasons, occurring between March to May and October to December, bring significant rainfall, with an annual average of around 1,200 millimeters (47 inches). During these periods, humidity levels can exceed 70%, and cloud cover is frequent, reducing sunshine hours to an average of 4-6 hours per day. In contrast, the dry season, lasting from June to February, is

marked by lower rainfall, clear skies, and higher temperatures. Relative humidity can drop to as low as 30%, and sunshine hours increase to an average of 7-8 hours per day. Despite the lower humidity, the intense sunlight and high UV index make it crucial to take precautions against sun exposure.

2.3 Temperatures

36°56'10.70"E experiences a tropical savanna climate, characterized by warm temperatures throughout the year with distinct wet and dry seasons. During the wet season, which typically falls between March and May, the area receives abundant rainfall, leading to lush vegetation and higher humidity levels. Temperatures during this period remain relatively warm, with average highs around 27°C (81°F) and lows around 18°C (64°F). The dry season, spanning from June to February, is marked by lower rainfall and drier conditions. Temperatures during this period can be quite hot, with average highs reaching 30°C (86°F) and lows around 15°C (59°F). However, the dry air and lower humidity often make the heat feel more tolerable compared to the humid wet season.

2.4 Precipitation

36°56'10.70"Ein Murang'a County experiences a bimodal rainfall pattern, characterized by two distinct rainy seasons. The primary rainy season occurs between March and May, during which the region receives the bulk of its annual rainfall. The secondary rainy season takes place between October and December, though it is generally less intense. On average, 36°56'10.70"E receives approximately 1,200 millimeters (47 inches) of rainfall annually. The highest rainfall amounts are typically recorded in April, with an average of around 230 millimeters (9 inches). November is the second wettest month, receiving an average of about 160 millimeters (6 inches) of rainfall. The driest months are January and February, with minimal rainfall during this period.

3. FIELD WORKS, TEST MATERIAL SAMPLING AND SAFETY PRECAUTIONS

3.1 Drilling of Exploratory Boreholes (Methodology)

The survey was carried out consistent with the requirements stipulated in the British Standard BS 5930:2015 (Code of Practice for Site Investigations) as a general guide.

Safety considerations in accordance with BS 6031:1981 clauses 4.1, 5.1.1 and clauses 5.1.2.1-3 were integrated into the strategy for use during the exploratory borehole drilling.

3.2 Equipment

CRANSFIELD MATERIAL TESTING CENTRE mobilized one rotary core drill rig, GY 150 Rig complete with 110mm nominal diameter casings and sampling tools for SPT and Bulk samples in soils and 86mm double tube swivel type core barrel with triplex piston water pump for diamond core drilling in rock with water as the flush media.

3.3 Standard Penetration Test

Standard Penetration Tests (SPT) was performed in the soil overburden encountered in the boreholes at 1.5 m intervals. The dynamic test entailed the driving of a split spoon sampler in to the soil layer at the bottom of the borehole by dropping a standard weight (63.5 kg), automatic trip hammer through a standard distance (760 mm). This was followed by recording of the number of blows (N) required to achieve 300 mm penetration (in 150 mm increments) after an initial seating drive of 150 mm or 25 blows, whichever was achieved first. The test termination criteria were as follows:

- When the full test drive (300 mm) was achieved, or;
- When a limiting blow count of 50 (N=50), also referred to as penetration 'refusal', was reached.

Based on data summarized by Skempton (1986), the generalized energy ratio (E_r) for our U.K. standard auto-trip hammer is 60%. The equation below can be used to acquire corrected N-values which is dependent on rod length ($C_r = 0.7$), sampler type ($C_s = 1.0$) and hole diameter ($C_d = 1.05$).

$$N_{corr} = N\frac{E_r}{60} \times 0.73$$

The depths at which the SPTs were carried out are given in the borehole logs. Full details of the blow counts and the SPT "N" value are given in borehole logs.

3.4 Dilatancy Correction Formula

This correction is used to adjust the SPT N-value when it is greater than 15. It recognizes that in saturated fine sands and silt soils, when the N-value is high, the soil's resistance to penetration can be significantly influenced by pore water pressures and the soil's dilatancy behavior.

The formula is: Ncorrected=15+0.5× (N–15)
Where:

- **Ncorrected** is the adjusted SPT N-value.
- **N** is the original (field) SPT N-value.
- If the original N-value is less than or equal to 15, no correction is applied.
- If the original N-value is greater than 15, the formula adjusts the value to better reflect the soil's actual behavior under loading conditions.

The correction formula of Ncorrected=15+0.5× (N–15) is specifically designed to adjust high SPT N-values for saturated fine sands and silts, ensuring that the results more accurately reflect the soil's behavior under loading conditions.

3.5 Conversion of SPT to Allowable bearing Capacities

To convert average SPT values to allowable bearing pressure, the relationship charts between SPT and Allowable bearing pressure developed by K. Terzaghi and R.B Peck were used, while limiting settlement of foundation to 25mm.

A commonly used empirical relationship is:

S_u= 6.5N (Principles of Foundation Engineering by Braja M. Das.)

Where:

- S_u = Undrained shear strength (kPa)

- N = Corrected SPT blow count

Using the S_u relationship you provided, we can estimate the ultimate bearing capacity and then apply a Factor of Safety (FS):

Ultimate Capacity (q_u): For a shallow footing, $q_u = 5.7 \times S_u$, as derived from Terzaghi's Bearing Capacity Theory (Terzaghi, 1943)

Safe Bearing Pressure (q_{safe}): Using a Factor of Safety of 3:

$$q_u = \frac{5.7 \ x \ Su}{3}$$

$$= \frac{5.7 \ x \ 6.5\mathbf{NCorr}}{3} = 12.35 \times \mathbf{N_{Corr}}$$

The SPT "N" values were then tabulated, and estimated bearing capacities computed based on the same as shown in in Table 2 below for ease of reference.

Table 2: SPT N Values and Bearing Capacity Estimation

Test Location	Depth (m)	SPT N-Value	SPT N Corr.-Value	Relative Density		Estimated Bearing Capacity Based on SPT N-Values Where S_L=25mm $q_a = 12.5N$ (KN/m²) Undrained shear strength, S_u (kPa)
				Undrained shear strength, S_u (kPa)	Consistency	
BH 1	**1.50**	9	9	59	STIFF	111
	3.00	11	11	72	STIFF	136
	4.50	14	14	91	STIFF	173
	6.00	16	16	104	STIFF	198
	7.50	19	17	111	VERY STIFF	210
	9.00	23	19	124	VERY STIFF	235
	10.50	25	20	130	VERY STIFF	247
	12.00	28	22	143	VERY STIFF	272
	13.50	30	23	150	VERY STIFF	284
	15.00	33	24	156	VERY STIFF	296
	16.50	34	25	163	VERY STIFF	309
	18.00	36	26	169	VERY STIFF	321
	19.50	38	27	176	VERY STIFF	333
	21.00	39	27	176	VERY STIFF	333
	22.50	40	28	182	VERY STIFF	346
	24.00	39	27	176	VERY STIFF	333
BH 2	**1.50**	7	7	46	FIRM	86

Test Location	Depth (m)	SPT N-Value	SPT N Corr.-Value	Relative Density		Estimated Bearing Capacity Based on SPT N-Values Where S_L=25mm q_a = 12.5N (KN/m²) Undrained shear strength, S_u (kPa)
				Undrained shear strength, S_u (kPa)	Consistency	
	3.00	10	10	65	STIFF	124
	4.50	13	13	85	STIFF	161
	6.00	17	16	104	STIFF	198
	7.50	21	18	117	VERY STIFF	222
	9.00	24	20	130	VERY STIFF	247
	10.50	25	20	130	VERY STIFF	247
	12.00	29	22	143	VERY STIFF	272
	13.50	32	24	156	VERY STIFF	296
	15.00	35	25	163	VERY STIFF	309
	16.50	37	26	169	VERY STIFF	321
	18.00	39	27	176	VERY STIFF	333
	19.50	39	27	176	VERY STIFF	333
	21.00	40	28	182	VERY STIFF	346
	22.50	42	29	189	VERY STIFF	358
	24.00	41	28	182	VERY STIFF	346

3.5 Material Sampling (Soil Sampling)

In this project, the following sampling techniques were employed:

Core sampling which included soils were obtained from the core barrels with the best samples being obtained by using double tube swivel type core barrels of 86mm diameter. Once the sample is acquired, the orientation of the sample was noted and then stored in to purpose made wooden core boxes ensuring that the depth orientation of each sample was maintained awaiting logging and analysis.

The typical core box consisted of five grooves, each groove with adequate dimensions (1.0 m length) for containing one meter of the core section at full recovery. Each core box was labeled to indicate the site name, borehole number, core runs and date of sampling. Photographs of the sample core boxes are provided in the appendices of this report.

3.6 Subsurface Conditions and Stratigraphy (Specific Geology)

In reference to the appended borehole logs (Appendix II). The descriptions of soil in the logs were derived from visual inspection methods conducted on-site by the supervising geologist, adhering to the guidelines specified in BS 5930:2015.

The stratigraphic boundaries indicated in the logs are inferred from intermittent sampling and observations of drilling resistance, which typically signify a transition from one layer to another. It is important to note that these boundaries should not be construed as precise planes of geological change but rather as approximations.

In order to facilitate geotechnical design, the stratigraphy summary provided below has been simplified focusing on the major strata. However, it is crucial to recognize that the information presented may not suffice for purposes of estimating subsurface material quantities across the site or for the associated stripping costs. Furthermore, the subsurface conditions confirmed at the boreholes will likely vary between and beyond the borehole locations, and additional review of the existing specific site geology will be necessary during excavation. In case the contractor or the engineer notices wide variations on site specific geology, they are at liberty to invite us for a joint review.

The lithology at the specific points of investigations as is evident in the borehole logs occurred as follows;

<u>BH No. 01</u>

- **0.0 – 25.0 meters:** Brown, medium grained GRAVELLY ELASTIC SILT

<u>BH No. 02</u>

- **0.0 – 25.0 meters:** Brown, medium grained GRAVELLY ELASTIC SILT WITH SAND

3.7 Measurement of Ground Water levels in boreholes

Ground water conditions in all the investigated boreholes were inspected during and upon completion of drilling, the water levels were monitored both in the morning before start of shift and in the evening after close of shift. However, no ground water was encountered in all the investigated boreholes.

3.8 Limitations of the Investigation Method

It should significantly be noted that the subsurface conditions are confirmed at the borehole locations only and that these conditions could vary in other areas of the site; between and beyond the boreholes. In addition, the boundaries between the various substrata, indicated on the borehole logs have been determined from non-continuous sampling. In this regard, these boundaries should be interpreted as inferred transitions rather than exact planes of geologic change.

4. LABORATORY MATERIAL TESTING

Representative fractions of disturbed samples recovered from the respective boreholes were obtained by quartering down respective samples using procedures defined in BS.1377 and the sub-samples were air dried afterwards to constant mass before commencing testing. The respective testing procedures and standards adopted laboratory analysis are outlined below in table No. 3.

Due to the occurrence of a consistent overburden soil layer with common descriptions as is evident from the borehole logs, soil material test was conducted over depth ranges as they occurred, and as illustrated below in sections 4.1 to 4.4 (ELASTIC SILT)

Table 3: Test Procedures and Standards

ITEM	TEST DESCRIPTIONS	PROCEDURES AND STANDARDS ADOPTED
A.	**Classification tests**	
A1.	Particle Size Distribution (Sieve analysis)	BS 1377-2:1990:9.2/9.3/9.4
A2.	Atterberg limits (Liquid, Plastic limit, Plasticity Index)	BS 1377-2:1990: 4.3: 1990
B.	**Compaction tests**	
B1.	Standard proctor tests- T99	BS 1377 PART- 4, 3.3: 1990
C	**Strength tests**	
C1.	Direct shear test	BS1377-7:1990, Clause 4
D.	**Calculations of Bearing capacities**	Principles of foundation engineering 6TH edition by Braja M. Das Braja, 2014, (Murthy, 2002)

4.1 Determination of Particle Size Distribution (Sieve Analysis)

Representative sub-samples of the materials obtained from the respective boreholes were subjected to particle size analysis using procedures described in BS.1377: Part 2:1990 Edition. The testing procedure was whereby the samples were soaked in a deflocculating agent solution for 24 hours and later washed over 0.075mm sieve to remove any adhering particles. The samples were then oven dried at 105 ± 5° C for a period of 24 ± 0.5 hours before sieving over a set of specified BS sieves to determine percentages of the sample passing each sieve.

Below in table No. 4 is a summary of the test results. Also attached in appendix I (10.1) of this report are the results in graphical form showing the particle size distribution.

Table 4: Particle size Distribution

SAMPLE ID	BS TEST SIEVES				Dominant Soil Type
	Depth(m)	Gravel (%>2.0mm)	Sand (2.0mm-0.075mm)	Fines (% <0.075mm)	
BH 01	0.0-25.0	22.8	5.3	71.9	GRAVELLY ELASTIC SILT(MH)
BH 02	0.0-25.0	19.2	16.2	64.6	GRAVELLY ELASTIC SILT WITH SAND (MH)

4.2 Determination of Atterberg limits

Both liquid and plastic limit tests were carried out on representative fractions of the samples recovered from the boreholes in order to establish the consistency of material encountered and with gradation, classification of the soil samples can be carried out. The procedure adopted was in accordance to BS.1377:1990 Edition.

Representative samples were prepared whereby the sub-samples were air dried and quartered down by riffling to produce representative fractions. Thereafter the material was passed through 0.425mm sieve to obtain sufficient material for testing. The recovered fractions were thoroughly mixed with de-aired water to a relatively stiff consistency and thereafter placed in an air tight container for 24 hrs. On

completion of the soaking period; the samples were placed on a glass plate where further increments of water were added to obtain the plastic limit, and liquid limit using the cone penetrometer method.

Based on the results obtained from respective samples, Unified Soil Classification System and AASHTO methods were used together with the particle size distribution to classify the soils.

Table No. 5 below outlines the test result which are also attached in graphical form in appendix I (10.2) of this report showing the consistency of the soil materials encountered.

Table 5: Atterberg limits

S/ID	Depth (M)	Liquid Limit (%)	Plastic Limit (%)	Plasticity Index (%)	Linear Shrinkage (%)	Soil classifications		
						USCS	Group Symbol	AASHTO
BH 01	0.0-25.0	93.04	53.7	39.34	14.29	GRAVELLY ELASTIC SILT	MH	A-7-5
BH 02	0.0-25.0	58.14	35.0	23.14	10.71	GRAVELLY ELASTIC SILT WITH SAND	MH	A-7-5

4.3 Determination of Maximum Dry Density/Optimum Moisture Content.

Representative samples of soil recovered from each of the boreholes were subjected to testing in accordance with procedures described in BS 1377: Part 4:1990 Edition.

Sub-samples were obtained by riffling the bulk samples for the purpose of obtaining respective fractions for testing. The material was air dried to a constant mass before obtaining from each sample five fractions each 2.5kg for testing. Predetermined amounts of water were added. The samples were then compacted in a standard mould using a 2.5 kg compacting effort to obtain the relationship between dry density and moisture content.

The maximum dry density and optimum moisture content as analyzed in overburden of the selected two boreholes are 1325Kg/m^3 to 1456Kg/m^3 and 37.6% and 27.3% respectively. The results are

presented below in table No. 6, also attached in appendix I are graphical representation for ease of reference.

Table 6: Dry density and Moisture content relationship

S/ID	Depth (M)	Maximum Dry Density (Kg/m^3)	Optimum Moisture Content (%)	Bulk Density (Kg/m^3)
BH 01	0.0-25.0	1325	37.6	1823
BH 02	0.00-25.0	1456	27.3	1853

4.4 Determination of Direct shear test

The shear strength of a soil is controlled by the effective stress that acts upon it.

Carried out in accordance with BS BS1377-7:1990, Clause 4 Sub-samples of the undisturbed and disturbed materials recovered from site were subjected to shear box test to evaluate their drained (effective) shear parameters: cohesion (c') and the angle of internal friction (φ').

The disturbed samples were remoulded at 95% MDD so as to simulate the condition at site as best as possible.

In a shear box apparatus, a specimen was subjected to a constant vertical load and horizontally sheared at various increments of vertical loading at a uniform rate such that the specimen is sheared in half along a failure plane guided by the two halves of the shear box. From the cohesion and the angle of shearing resistance values obtained, the bearing capacity was calculated.

Subsequently, a summary of the direct shear stress parameters **Ø** and **C** are presented in table No. 7 below, and also attached in graphical representation in appendix I of this report.

Table 7: Direct shear strength

S/ID	Depth (M)	Bulk Density (Kg/m^3)	Angle of Internal friction; Ø (°)	Cohesion C (Kpa)
BH 01	0.0-25.0	1823	17	10
BH 02	0.0-25.0	1853	16	12

5. CALCULATION/ESTIMATION OF BEARING CAPACITIES IN SOIL FORMATIONS

The bearing capacities have been computed from principles of foundation engineering 6th edition by braja m.das using direct shear test parameters and terzhagi's bearing capacity constants illustrated in tables no. 8, 9 and 10 below.

$Q_{ult} = 1.3\ Cu\ Nc + \Upsilon\ D\ Nq + 0.4\ \Upsilon\ B\ N\Upsilon$

Where;

Q_{ult} is the Gross Ultimate Bearing Capacity

C_u is the cohesion of soil

Υ is the bulk density/unit weight of soil

B is the width of the footing

D is the depth of the footing

N_C, Nq and N_Υ are Terzhagi's Constants, subject angle of Internal Friction; Ø obtained during direct shear test.

Based on the results of analysis using Terzaghi's bearing capacity theory, it is evident that the allowable bearing pressure increases with both the depth of embedment and the width of the foundation base. This is primarily due to the increase in overburden pressure and the greater mobilization of shear resistance in the supporting soil mass.

We have therefore provided **tables 8 and 9** showing the calculated bearing pressures for foundation width (B) ranging from **1.0 to 4.0 meters,** considering overburden conditions of 1.5 meters and 2.0 meters. These values serve as a guide for selecting the appropriate foundation depth and width that ensures this is for depth between 1.5m to 25.0m on both boreholes.

Table 8: Bearing capacities computed from direct shear test parameters (Overburden of 1.5m)

S/ID	Base Width (M)	ULTIMATE BEARING CAPACITIES (KN/m²)	SAFE BEARING CAPACITIES (KN/m²)
BH 1	1.0	348	116
	2.0	363	121
	3.0	378	126
	4.0	393	131
BH 2	1.0	357	119
	2.0	369	123
	3.0	384	128
	4.0	396	132

Table 9: Bearing capacities computed from direct shear test parameters (Overburden of 2.0m)

S/ID	Base Width (M)	ULTIMATE BEARING CAPACITIES (KN/m²)	SAFE BEARING CAPACITIES (KN/m²)
BH 1	1.0	396	132
	2.0	411	137
	3.0	426	142
	4.0	444	148
BH 2	1.0	402	134
	2.0	414	138
	3.0	429	143
	4.0	441	147

Table 10: Terzhagi's constants

PRINCIPLES OF FOUNDATION ENGINEERING 6TH EDITION BY BRAJA M. DAS TERZHAGI'S BEARING CAPACITY CONSTANTS							
Angles	Corresponding Constants				Angles	Corresponding Constants	
Ø	N_C	N_q	N_{ya}		Ø	N_C	N_q	N_{ya}
0	5.7	1	0		26	27.09	14.21	9.84
1	6	1.1	0.01		27	29.24	15.9	11.6
2	6.3	1.22	0.04		28	31.61	17.81	13.7
3	6.62	1.35	0.06		29	34.24	19.98	16.18
4	6.97	1.49	0.1		30	37.16	22.46	19.13

PRINCIPLES OF FOUNDATION ENGINEERING 6TH EDITION BY BRAJA M. DAS TERZHAGI'S BEARING CAPACITY CONSTANTS								
5	7.34	1.64	0.14		31	40.41	25.28	22.65
6	7.73	1.81	0.2		32	44.04	28.52	26.87
7	8.15	2.0	0.27		33	48.09	32.23	31.94
8	8.6	2.21	0.35		34	52.64	36.5	38.04
9	9.09	2.44	0.44		35	57.75	41.44	45.41
10	9.61	2.69	0.56		36	63.53	47.16	54.36
11	10.16	2.98	0.69		37	70.01	53.8	65.27
12	10.76	3.29	0.85		38	77.5	61.55	78.61
13	11.41	3.63	1.04		39	85.97	70.61	95.03
14	12.11	4.02	1.26		40	95.66	81.27	115.31
15	12.86	4.45	1.52		41	106.81	93.85	140.51
16	13.68	4.92	1.82		42	119.67	108.75	171.99
17	14.6	5.45	2.18		43	134.58	126.5	211.56
18	15.12	6.04	2.59		44	151.95	147.74	261.6
19	16.56	6.7	3.07		45	172.28	173.28	325.34
20	17.69	7.44	3.64		46	196.22	204.19	407.11
21	18.92	8.26	4.31		47	224.55	241.8	512.84
22	20.27	9.19	5.09		48	258.28	287.85	650.67
23	21.75	10.23	6		49	298.71	344.63	831.99
24	23.36	11.4	7.08		50	347.5	415.14	1072.8
25	25.13	12.72	8.34					

6. DISCUSSIONS AND CONCLUSIONS

The following are based on the factual data obtained from this investigation coupled with those from the laboratory analysis as above, and are intended for use by the **STATE DEPARTMENT FOR HOUSING AND URBAN DEVELOPMENT** and its consultants. Contractors bidding or providing services on this project should review the factual data and determine their own conclusions regarding construction methods and scheduling.

i The exploratory borings exhibit a layer of brown, medium grained GRAVELLY ELASTIC SILT extending from 0.0 – 25.0 meters at the end of the run for BOREHOLE 1.

For BOREHOLE 2, the exploratory boring exhibit a layer of Brown, medium grained GRAVELLY ELASTIC SILT WITH SAND extending from 0.0 – 25.0 meters at the end of the run.

ii From particle size distribution and Atterberg limits, and using the USCS and AASHTO soil classification systems, these overburden soils classify majorly as GRAVELLY ELASTIC SILT and GRAVELLY ELASTIC SILT WITH SAND (MH); A-7-5. In accordance with AASHTO ASTM D3282-09, Table 2, these soils classify as fair to poor subgrade materials.

iii The structural engineer shall take note of this overburden layer with varying bearing pressures at different depths from **0.086Kpa** to **0.333Kpa** as illustrated in tables No. 2, 8 and 9 (SPT and Direct Shear box test)

iv No underground water was struck at any boreholes.

v The onsite material which falls under A-7-5 (AASHTO) as in (ii) are considered poor/fair for subgrade or backfill material is not fit for backfill during construction. In this regard, the project engineers shall consider use of foreign approved material for backfill.

7. RECOMMENDATIONS

7.1 Slab-On-Grade Drainage

All ground slabs must be provided with a capillary moisture barrier and drainage layer. The capillary moisture barrier can be made by placing the slab on a minimum 200 mm layer of clear 19 mm stone compacted by vibration to a dense state. This stone serves also as the drainage media for the subfloor drainage system, which is required. Any aggregate fill placed beneath the floor slab must be compacted to not less than 98 % of Standard Proctor maximum dry density, and moisture contents ranging from 85% to 105%.

7.2 Foundation type and allowable bearing pressure

i With prior knowledge from the client's brief in this project, we, in this regard recommend deep foundation type using reinforced concrete isolated, strap and combined footings or raft foundation as per the working load and foundation column spacing

ii The soil overburden at between approximately 0.0 - 25.0m may be susceptible to volume change, expansion with varying moisture content and has concentration of relatively low bearing pressure, this stratum is not recommended to bed foundation layer for heavy structures, and such foundations should not be placed on the same. However, when foundation is laid on them, we recommend the following:

1. **Foundation Depth:** Ensure that the foundation is placed below the zone of seasonal moisture variation soil. This is depth below **2.0m** below the existing ground.
2. **Drainage:** Implement proper site drainage to prevent water from accumulating near the foundation, which could exacerbate the loss of soil shear strength.
3. **Material Selection:** Consider using moisture-resistant materials for the foundation, we recommend incorporating two layers of 1000 gauge polythene sheet (or an approved equivalent) beneath the foundation slab as a moisture barrier. Additionally, ensure proper sub-base compaction and provide a well-drained subgrade to minimize water retention.

4. **Monitoring:** We Recommend that the engineer closely monitor the soil moisture content during construction to avoid unexpected moisture variations.

iii The design engineers shall use as a guide the obtained range of safe bearing pressures at various depths as per the laboratory analysis, field investigation, engineering judgments; tables 2, 8 and 9 (GRAVELLY ELASTIC SILT and GRAVELLY ELASTIC SILT WITH SAND) for characteristic service vertical load to calculate the foundation size.

iv An effective storm water drainage system should be installed on the site to lead away all surface/ roof water away from the foundation areas without allowing any water ponding. The backfill of the foundation sides should be of selected material e.g. lateritic firm soil that is compacted adequately to Maximum dry density at Optimum Moisture Content to form an impermeable layer to prevent ingress of water into the foundation layers

v The results of our analysis, conducted in accordance with Terzaghi's bearing capacity theory, indicate that the allowable bearing pressure increases with both the depth of overburden and the width of the foundation base. This is attributed to the increase in effective overburden pressure and the enhanced mobilization of shear strength in the bearing stratum.

vi We recommend that foundation to be set at depths ranging from **2.0 to 5.0 meters** below the existing ground level, within the **gravelly elastic silt and gravelly elastic silt with sand stratum**. Considering an overburden depth of **2.0 meters** above the **foundation base,** a safe bearing capacity of **120 kN/m²** is recommended. However, the engineer is at liberty to adjust the foundation depth and base size, provided the selected depth and base size aligns with the safe bearing pressures presented in the illustrated table above.

vii Foundations shall be designed as per British standard or Euro code relevant sections.

viii **50mm** lean concrete must be used before casting the foundation base. The lean concrete must be place on stable and compacted ground with the approval of the geotechnical engineer.

ix Height difference between the nearby foundation bases shall be less than 40% the free distance between them to avoid foundation stress overlap.

x The consulting structural Engineer shall be invited for supervision before casting the foundation bases or rafts.

xi To avoid differential settlement between the nearby foundations, all possible means shall be implemented to have more or less uniform foundation stresses.

7.3 Earth work fill and earth work cut

i. Cut surfaces shall not be steeper than works 1.6H: 1V

ii. If there exist building adjacent to the excavation it shall be protected with retaining structures such as piles etc.

iii. Fill material shall be gravel with maximum 15% fine pass #200 and CBR not less than 14, T99.

iv. Maximum layer for compaction of the fill shall be 200mm

8. LIMITATIONS AND RISKS

This geotechnical examination has been carried out using investigation techniques and engineering analysis methods consistent with those ordinarily exercised by CRANSFIELD MATERIAL TESTING CENTRE and other engineering practitioners, working under similar conditions and subject to the time, financial and physical constraints applicable to this project. The discussions and recommendations that have been presented are based on the factual data obtained from this investigation.

It must be recognized that there are special risks whenever engineering or related disciplines are applied to identify subsurface conditions. A comprehensive sampling and testing programme implemented in accordance with the most stringent level of care may fail to detect certain conditions. CRANSFIELD MATERIAL TESTING CENTRE has assumed for the purposes of providing design parameters and advice that the conditions that exist proximal to the sampling point are similar to those found at the sample location. These conditions may differ from those that actually exist. It may not be possible to drill sufficient number of boreholes or sample and report them in a way that would provide all the subsurface information that could affect construction costs, techniques, equipment and scheduling. Contractors and quantity Surveyors bidding on or undertaking work on this project should be directed to draw their own conclusions as to how the subsurface conditions may affect them, based on their own investigations and their own interpretations of the factual investigation results, cognizant of the risks implicit in the subsurface investigation activities.

It must equally be noted that the passage of time, natural occurrences, and direct or indirect human intervention at or near the site have the potential to alter subsurface conditions.

The design parameters provided and the engineering discussion are based on the factual data obtained from this investigation and visual observations made at the site by CRANSFIELD MATERIAL TESTING CENTRE, and are intended for use by owner and its retained designers in the design phase of the project. If there are changes to the project scope and development features, the interpretations made from the subsurface information, the geotechnical design parameters and comments relating to

constructability issues and quality control may not be relevant to the revised project.

The bearing capacities were computed from soil direct shear from principles of soil mechanics and Standard Handbook for Civil Engineers, 5th Edition, and Section 7.10

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STATE DEPARTMENT FOR HOUSING AND URBAN DEVELOPMENT and its retained design consultants are the authorized users.

Finally, this completes the brief outlined in the scope of works; however, should you require clarification or, indeed, any assistance of the data presented therein, please do not hesitate in contacting us.

9. REFERENCES

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Sign…….

Checked by;

Eng. Joel Wairua

Sign…… Official Stamp

CRANSFIELD MATERIAL TESTING
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Sign:.....
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10. Appendix I (LABORATORY AND INSITU TEST REPORTS)

10.1 Determination of Particle Size Distribution

Cransfield Materials Testing Center

Quality Assurance

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Email: info@cmtc.co.ke
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PARTICLE SIZE DISTRIBUTION

BS 1377 2:1990: 9.2/9.3/9.4 & 9.5

Client:	STATE DEPARTMENT FOR HOUSING AND URBAN DEVELOPMENT					**Date submitted:**	9-Jun-26
Project/Site Name:	PROPOSED CONSTRUCTION OF KIRIAINI AHP IN MATHIOYA CONSTITUENCY, MURANG'A COUNTY WITH ASSOCIATED AMENITIES AND INFRASTRUCTURE					**Date tested:**	11-Jun-26
Sample ID:	BH1	**Sample No.:**	-	**Sample depth (M):**	0.0-25.0	**Sampled and Submitted by:**	Cransfield

Sample preparation details		**Moisture content at sample preparations**		
Initial quatered and air dried mass (gm)	328.0	**Wet weight of sample (gm)**		186.5
Washed and oven dried mass (gm)	245.4	**Dry weight of sample (gm)**		149.0
Fine mass (gm)	176.4	**Wet weight of water (gm)**		37.5
Fines percent (%)	71.9	**Moisture content (%)**		25.2

Soil Classification					
GRAVEL (%):	22.8	**CLAY/SILT (%):**	71.9	**USCS**	**GRAVELLY ELASTIC SILT(MH)**
SAND (%):	5.3			**AASHTO**	**A-7-5**

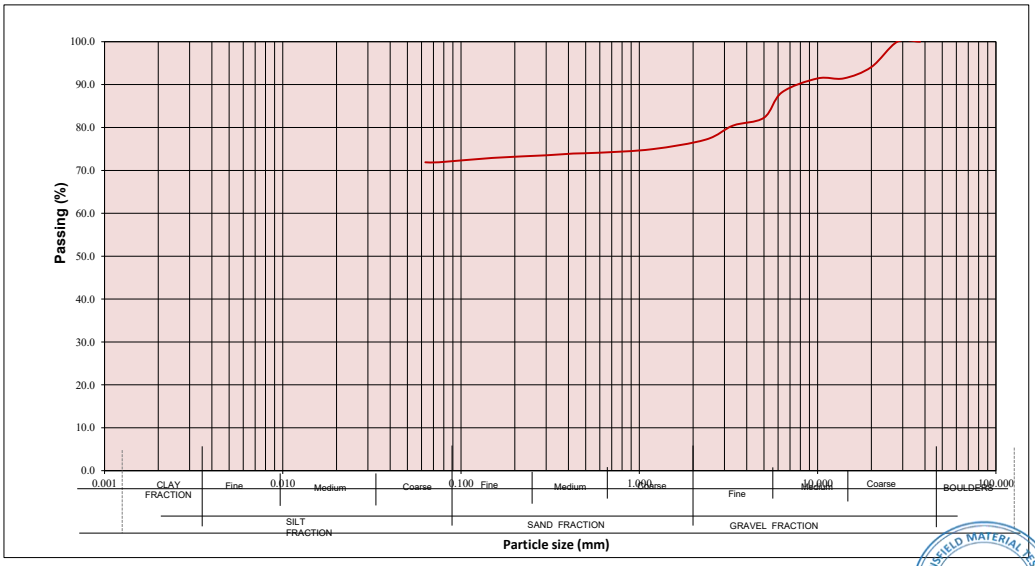
Sieve size (mm)	**Retained mass (gm)**	**% Retained (%)**	**Cumulative passing percentage (%**
63	0.0	0.0	100
50	0.0	0.0	100.0
37.5	0.0	0.0	100.0
28	0.0	0.0	100.0
20	14.5	5.9	94.1
14	6.5	2.6	91.4
10	8.0	3.3	88.2
6.3	8.0	3.3	84.9
5	6.5	2.6	82.3
3.35	4.5	1.8	80.4
2.36	8.0	3.3	77.2
1.18	5.5	2.2	74.9
0.6	2.0	0.8	74.1
0.425	0.5	0.2	73.9
0.3	1.0	0.4	73.5
0.15	1.5	0.6	72.9
0.075	2.5	1.0	71.9
0.063	0.0	0.0	71.9
<0.063	176.4	71.9	-
<0.075		-	-
TOTAL	69.0	-	-

Hydrometer Analysis to BS 1377-2:1990:9.5

Dry Weight	**(g)**	-	**Hydrometer Type**		-
S.G	**(Mg/m^3)**	-	**Zero Correction factor**		-
Temprature	**(C^O)**	-	**S.G Correction factor**		-
K Factor		-	**Temperature Correction Factor**		-

Time, min	**Actual HR rh**	**Adjusted HR(Rh)**	**Composite correction**	**Corrected HR**	**Effective depth (cm)**	**Diameter; D Kv(L/T)**	**% Fines in Suspension**	**% Fines based on whole sample**
0.25	-	-	-	-	-	-	-	-
0.50	-	-	-	-	-	-	-	-
1	-	-	-	-	-	-	-	-
2	-	-	-	-	-	-	-	-
4	-	-	-	-	-	-	-	-
8	-	-	-	-	-	-	-	-
15	-	-	-	-	-	-	-	-
30	-	-	-	-	-	-	-	-
60	-	-	-	-	-	-	-	-
120	-	-	-	-	-	-	-	-
240	-	-	-	-	-	-	-	-
480	-	-	-	-	-	-	-	-
1440	-	-	-	-	-	-	-	-

PARTICLE SIZE DISTRIBUTION GRAPH



Tested by:	JILLO	**Date reported:**	13-Jun-26	**Checked by:**

Cransfield Materials Testing Center

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Website: www.cmtc.co.ke

PARTICLE SIZE DISTRIBUTION

BS 1377 2:1990: 9.2/9.3/9.4 & 9.5

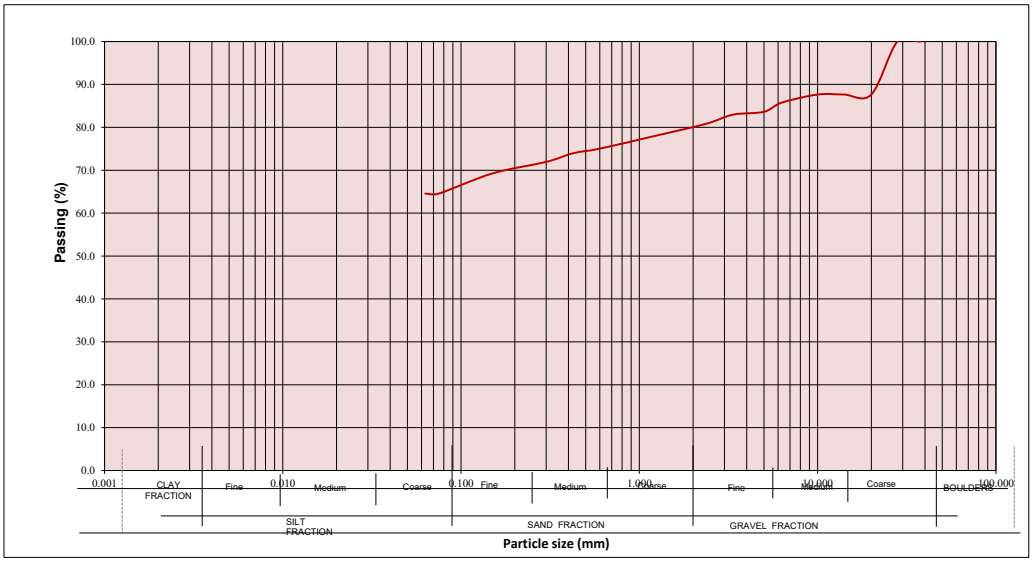
Client:	STATE DEPARTMENT FOR HOUSING AND URBAN DEVELOPMENT					**Date submitted:**	9-Jun-26
Project/Site Name:	PROPOSED CONSTRUCTION OF KIRIAINI AHP IN MATHIOYA CONSTITUENCY, MURANG'A COUNTY WITH ASSOCIATED AMENITIES AND INFRASTRUCTURE					**Date tested:**	11-Jun-26
Sample ID:	BH2	**Sample No.:**	-	**Sample depth (M):**	0.0-25.0	**Sampled and Submitted by:**	Cransfield

Sample preparation details			**Moisture content at sample preparations**		
Initial quatered and air dried mass	(gm)	422.0	Wet weight of sample	(gm)	119.0
Washed and oven dried mass	(gm)	372.5	Dry weight of sample	(gm)	106.5
Fine mass	(gm)	240.5	Wet weight of water	(gm)	12.5
Fines percent	(%)	64.6	Moisture content	(%)	11.7

Soil Classification					
GRAVEL (%):	19.2	**CLAY/SILT (%):**	64.6	**USCS**	**GRAVELLEY ELASTIC SILT WITH SAND (MH)**
SAND (%):	16.2			**AASHTO**	**A-7-5**

Wet & Dry Sieve Analysis to BS 1377 2:1990: 9.2/9.3/9.4				**Hydrometer Analysis to BS 1377-2:1990:9.5**									
Sieve size (mm)	**Retained mass (gm)**	**% Retained (%)**	**Cumulative passing percentage (%**	**Dry Weight**	**(g)**	-	**Hydrometer Type**	-					
63	0.0	0.0	100	**S.G**	**(Mg/m^3)**	-	**Zero Correction factor**	-					
50	0.0	0.0	100.0	**Temprature**	**(C^o)**	-	**S.G Correction factor**	-					
37.5	0.00	0.0	100.0	**K Factor**		-	**Temperature Correction Factor**	-					
28	46.00	12.4	87.6										
20	0.00	0.0	87.6										
14	0.00	0.0	87.6	**Time, min**	**Actual HR rh**	**Adjusted HR(Rh)**	**Composite correction**	**Corrected HR**	**Effective depth (cm)**	**Diameter; D Kv(L/T)**	**% Fines in Suspension**	**% Fines based on whole sample**	
10	7.00	1.9	85.8										
6.3	4.50	1.2	84.6										
5	3.50	0.9	83.6	0.25	-	-	-	-	-	-	-	-	
3.35	2.50	0.7	83.0	0.50	-	-	-	-	-	-	-	-	
2.36	8.00	2.1	80.8	1	-	-	-	-	-	-	-	-	
1.18	11.00	3.0	77.9	2	-	-	-	-	-	-	-	-	
0.6	10.50	2.8	75.0	4	-	-	-	-	-	-	-	-	
0.425	4.00	1.1	74.0	8	-	-	-	-	-	-	-	-	
0.3	7.50	2.0	71.9	15	-	-	-	-	-	-	-	-	
0.15	10.00	2.7	69.3	30	-	-	-	-	-	-	-	-	
0.075	17.50	4.7	64.6	60	-	-	-	-	-	-	-	-	
0.063	0.0	0.0	64.6	120	-	-	-	-	-	-	-	-	
<0.063	240.5	64.6	-	240	-	-	-	-	-	-	-	-	
<0.075		-	-	480	-	-	-	-	-	-	-	-	
TOTAL	132.0	-	-	1440	-	-	-	-	-	-	-	-	

PARTICLE SIZE DISTRIBUTION GRAPH



Tested by:	JILLO	**Date reported:**	13-Jun-26	**Checked by:**

10.2 Determination of Atterberg limits

Cransfield Materials Testing Center
Quality Assurance

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Tel: +254 703410973/ +254 721740587
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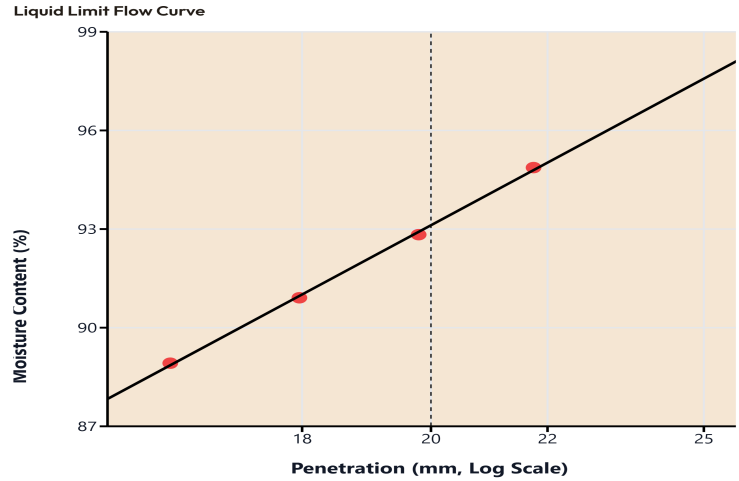
ATTERBERG LIMITS (BS 1377 PART 2, 4.3 : 1990)

Client name: AHP		
Project/Site name: KIRIANI AFFORDABLE HOUSING PROJECT		
Sampled by: -	**Date submitted:** -	**Date tested:** -
Sample ID: BHO1	**Sample depth:** -	**Sample No:** 0.0-25.0

ATTERBERG LIMITS TEST

	Trial 1 (LL) (HCL)	Trial 2 (LL) (WK)	Trial 3 (LL) (2KZ)	Trial 4 (LL) (SH)	Trial 1 (PL) (PN)	Trial 2 (PL) (PKX)
Container No	HCL	WK	2KZ	SH	PN	PKX
Penetration (mm)	16.16	17.96	19.8	21.75	-	-
Cont + Wet Soil (g)	23.54	22.26	25.8	22.8	10.16	10.22
Cont + Dry Soil (g)	12.46	11.66	13.38	11.7	6.64	6.62
Container (g)	0	0	0	0	0	0
Wt Moisture (g)	11.08	10.6	12.42	11.1	3.52	3.6
Wt Dry Soil (g)	12.46	11.66	13.38	11.7	6.64	6.62
Moisture Content (%)	**88.92**	**90.91**	**92.83**	**94.87**	**53.01**	**54.38**

PLASTIC LIMIT **53.7**



LINEAR SHRINKAGE

Initial length (mm)	140
Final length (mm)	120
Shrinkage (%)	14.29

RESULTS SUMMARY

LIQUID LIMIT (%)	93.04
PLASTIC LIMIT (%)	53.7
PLASTICITY INDEX (%)	39.34
Passing 425 µm (%)	73.9
MODULUS OF PLASTICITY	2907.23
LINEAR SHRINKAGE (%)	14.29

SOIL CLASSIFICATION

USCS	Silt of High Plasticity	**MH**
AASHTO	**A-7-5**	Elastic silty soil

Tested by: JILLO **Date reported:** _____ **Checked by:** _____

Cransfield Materials Testing Center
Quality Assurance

Cransfield Materials Testing Center
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Website: www.cmtc.co.ke

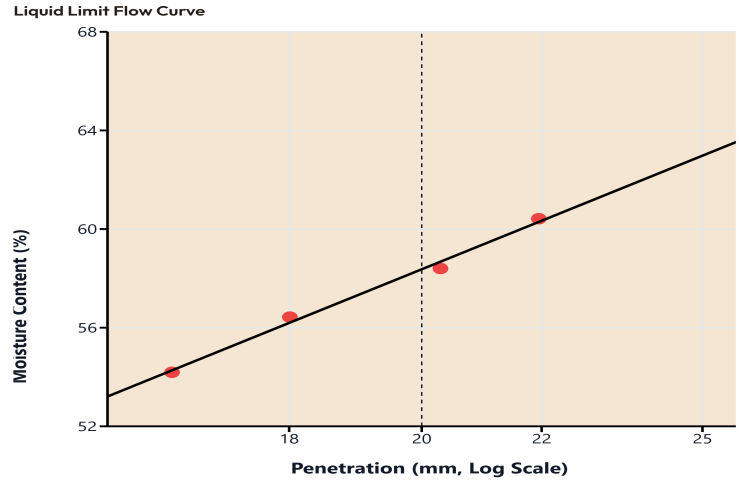
ATTERBERG LIMITS (BS 1377 PART 2, 4.3 : 1990)

Client name: AHP		
Project/Site name: KIRIANI AFFORDABLE HOUSING PROJECT		
Sampled by: -	**Date submitted:** -	**Date tested:** -
Sample ID: BH02	**Sample depth:** -	**Sample No:** 0.0-25.0

ATTERBERG LIMITS TEST

	Trial 1 (LL) (XG)	Trial 2 (LL) (OF)	Trial 3 (LL) (GH)	Trial 4 (LL) (OP)	Trial 1 (PL) (MW)	Trial 2 (PL) (BE)
Container No	XG	OF	GH	OP	MW	BE
Penetration (mm)	16.4	18.01	20.3	21.95	-	-
Cont + Wet Soil (g)	25.75	26.5	24.6	26.18	11.12	11.21
Cont + Dry Soil (g)	16.7	16.94	15.53	16.32	8.26	8.28
Container (g)	0	0	0	0	0	0
Wt Moisture (g)	9.05	9.56	9.07	9.86	2.86	2.93
Wt Dry Soil (g)	16.7	16.94	15.53	16.32	8.26	8.28
Moisture Content (%)	**54.19**	**56.43**	**58.4**	**60.42**	**34.62**	**35.39**

PLASTIC LIMIT **35**



LINEAR SHRINKAGE

Initial length (mm)	140
Final length (mm)	125
Shrinkage (%)	10.71

RESULTS SUMMARY

LIQUID LIMIT (%)	58.14
PLASTIC LIMIT (%)	35
PLASTICITY INDEX (%)	23.14
Passing 425 µm (%)	74
MODULUS OF PLASTICITY	1712.36
LINEAR SHRINKAGE (%)	10.71

SOIL CLASSIFICATION

USCS	Silt of High Plasticity	**MH**
AASHTO	**A-7-5**	Elastic silty soil

Tested by: JILLO **Date reported:** _____ **Checked by:** _____

CRANSFIELD MATERIAL TESTING
P.O. BOX 59401-00100 NAIROBI

10.3 Dry Density Moisture Content relationships

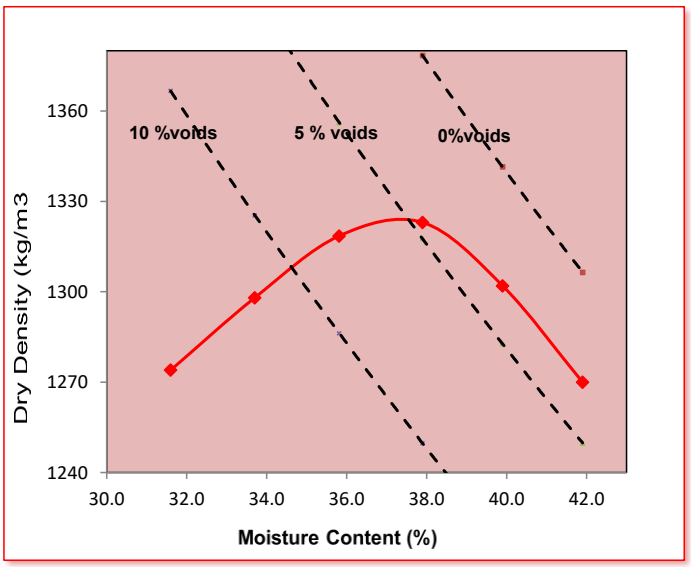
Cransfield Materials Testing Center
Quality Assurance

Cransfield Materials Testing Center
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Nairobi, Kenya
Runda Grove House No. 278
Tel: +254 703410973/ +254 733725775
Email: info@cmtc.co.ke
Website: www.cmtc.co.ke

DENSITY/MOISTURE CONTENT RELATIONSHIP (BS 1377 PART- 4, 3.3 : 1990)

Client name:	STATE DEPARTMENT FOR HOUSING AND URBAN DEVELOPMENT					
Project/Site Name:	PROPOSED CONSTRUCTION OF KIRIAINI AHP IN MATHIOYA CONSTITUENCY, MURANG'A COUNTY WITH ASSOCIATED AMENITIES AND INFRASTRUCTURE					
Sampled and submitted by:	CRANSFIELD	**Date submitted:**	10-Jun-26	**Date tested:**	13-Jun-26	
Sample ID:	BH 1	**Sample depth (M):**	0.0-25.0	**Location:**		

Moisture addition	PMC	50CC	200CC	250CC	300CC	350CC
Wt of mould + wet material (g)	4564	4617	4667	4697	4695	4677
Wt of mould + Base (g)	3050	3050	3050	3050	3050	3050
Wt wet material (g)	1514	1567	1617	1647	1645	1627
Volume of mould (m^3)	0.903	0.903	0.903	0.903	0.903	0.903
Bulk density (kg/m^3)	**1677**	**1735**	**1791**	**1824**	**1821**	**1802**
Container No	DT	6Y	FD	JI	JI	OL
Wt of container + wet material (g)	204.4	169.0	201.0	201.2	204.1	256.3
Wt dry material (g)	155.3	126.4	148.0	145.9	145.9	180.6
Wt of moisture (g)	49.07	42.60	53.00	55.30	58.21	75.67
Wt of container (g)	-	-	-	-	-	-
Moisture content (%)	**31.6**	**33.7**	**35.8**	**37.9**	**39.9**	**41.9**
Dry density (kg/m^3)	**1274**	**1298**	**1319**	**1323**	**1302**	**1270**



Maximum Dry Density (kg/m^3):	**1325**	**Optimum Moisture Content (%):**	**37.6**
Bulk Density (kg/m^3):	**1823**		

Tested by:	CRANSFIELD	**Date reported:**	14-Jun-26	**Checked by:**	

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Cransfield Materials Testing Center
Quality Assurance

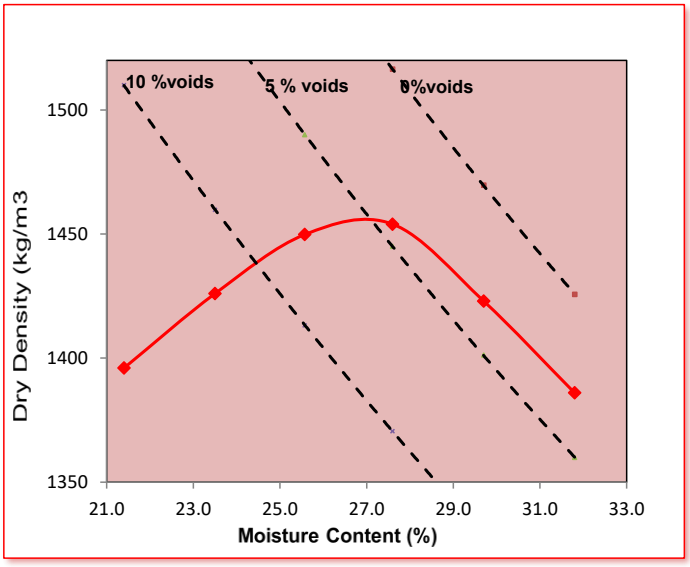
Cransfield Materials Testing Center
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Email: info@cmtc.co.ke
Website: www.cmtc.co.ke

DENSITY/MOISTURE CONTENT RELATIONSHIP (BS 1377 PART- 4, 3.3 : 1990)

Client name:	STATE DEPARTMENT FOR HOUSING AND URBAN DEVELOPMENT					
Project/Site Name:	PROPOSED CONSTRUCTION OF KIRIAINI AHP IN MATHIOYA CONSTITUENCY, MURANG'A COUNTY WITH ASSOCIATED AMENITIES AND INFRASTRUCTURE					
Sampled and submitted by:	CRANSFIELD	**Date submitted:**	10-Jun-26	**Date tested:**	13-Jun-26	
Sample ID:	BH02	**Sample depth (M):**	0.00-25.0	**Location:**		

Moisture addition	PMC	250CC	300CC	350CC	400CC	450CC
Wt of mould + wet material (g)	4580	4640	4694	4725	4717	4700
Wt of mould + Base (g)	3050	3050	3050	3050	3050	3050
Wt wet material (g)	1530	1590	1644	1675	1667	1650
Volume of mould (m^3)	0.903	0.903	0.903	0.903	0.903	0.903
Bulk density (kg/m^3)	**1695**	**1761**	**1821**	**1855**	**1846**	**1827**

Container No	DT	6Y	DA	JI	JI	OL
Wt of container + wet material (g)	188.5	156.1	191.5	186.2	189.2	238.0
Wt dry material (g)	155.3	126.4	152.5	145.9	145.9	180.6
Wt of moisture (g)	33.23	29.70	39.00	40.27	43.33	57.43
Wt of container (g)	-	-	-	-	-	-
Moisture content (%)	**21.4**	**23.5**	**25.6**	**27.6**	**29.7**	**31.8**
Dry density (kg/m^3)	**1396**	**1426**	**1450**	**1454**	**1423**	**1386**



Maximum Dry Density (kg/m^3):	**1456**	**Optimum Moisture Content (%):**	**27.3**
Bulk Density (kg/m^3):	**1853**		

Tested by:	CRANSFIELD	**Date reported:**	14-Jun-26	**Checked by:**	

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Sign:

10.4 Direct Shear Box Test

UNCONSOLIATED DIRECT SHEAR TEST (BS1377-7:1990,4)

STATE DEPARTMENT FOR HOUSING AND URBAN DEVELOPMENT

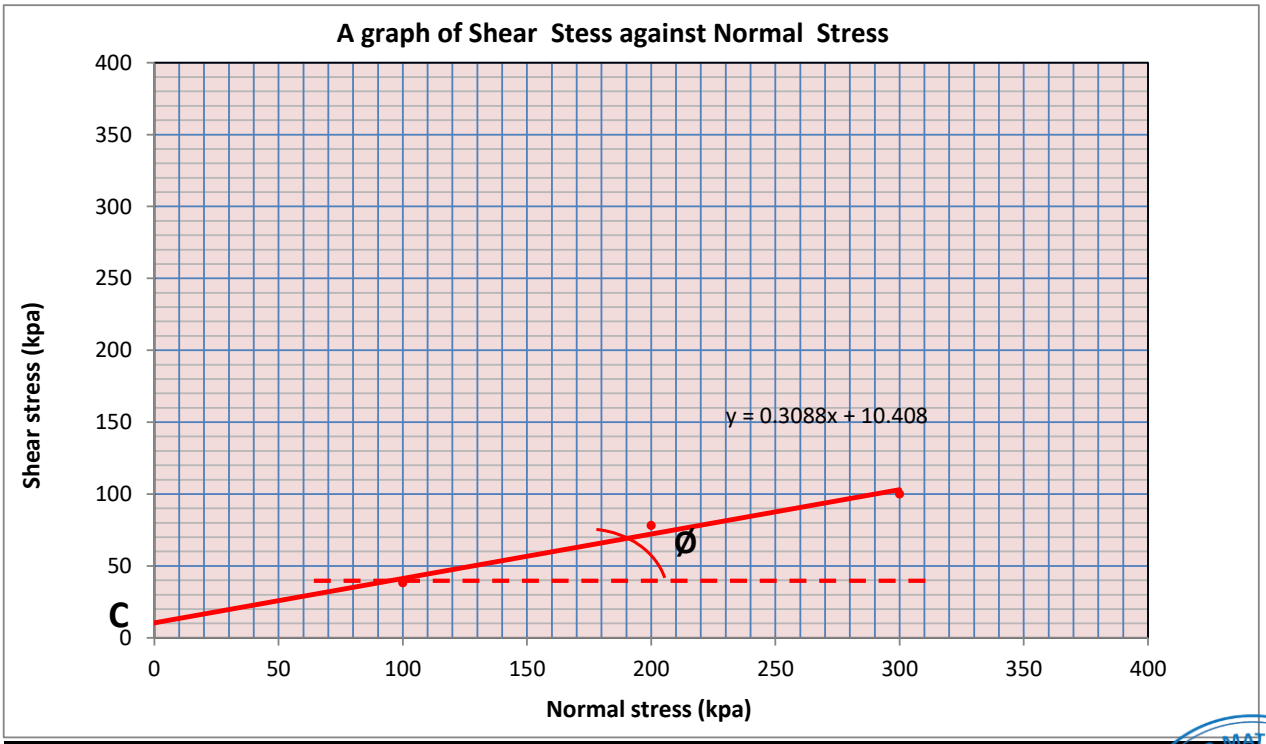
PROPOSED CONSTRUCTION OF KIRIAINI AHP IN MATHIOYA CONSTITUENCY, MURANG'A COUNTY WITH ASSOCIATED AMENITIES AND INFRASTRUCTURE

Sampled and submitted by:	CRANSFIELD	**Date submitted:**	10-Jun-26	**Date tested:**	12-Jun-26
Sample ID:	BH 01	**Sample depth (M):**	1.5	**Sample No:**	_

SPECIMEN AND TEST DETAILS

Sample Length; L (cm):	6.00	**Natural Moisture Content (%):**	37.6
Sample height (cm):	2.50	**Bulk Density (g/m^3):**	1823
Sample Area; A_o (cm^2):	36.00	**Maximum Dry Density (g/m^3):**	1325
Sample Volume (cm^3):	90.00	**Mass of specimen in shear ring (g):**	164.09
Shear Rate (mm/min):		**Specific gravity - Assumed (g/m^3):**	2.60

Max. load cell Reading (KN)	0.121 0.247 0.314	**Specimen Displacement (cm)**	0.733 0.733 0.769	**Corrected Specimen Area (cm^2)**	31.60 31.60 31.39	**Normal Stress (kpa):**	100 200 300	**Shear stress (kpa):**	38.29 78.16 100.04



C	10 **kpa**
Ø	17 °
SBP	**116 KN/m2**

Tested by:	PAUL	**Date Reported:**	14-Jun-26	**Checked by:**

CRANSFIELD MATERIAL TESTING
Email: c.materialcenter@gmail.com
Sign:
P.O. Box 59401-00100 NAIROBI

UNCONSOLIATED DIRECT SHEAR TEST (BS1377-7:1990,4)

STATE DEPARTMENT FOR HOUSING AND URBAN DEVELOPMENT

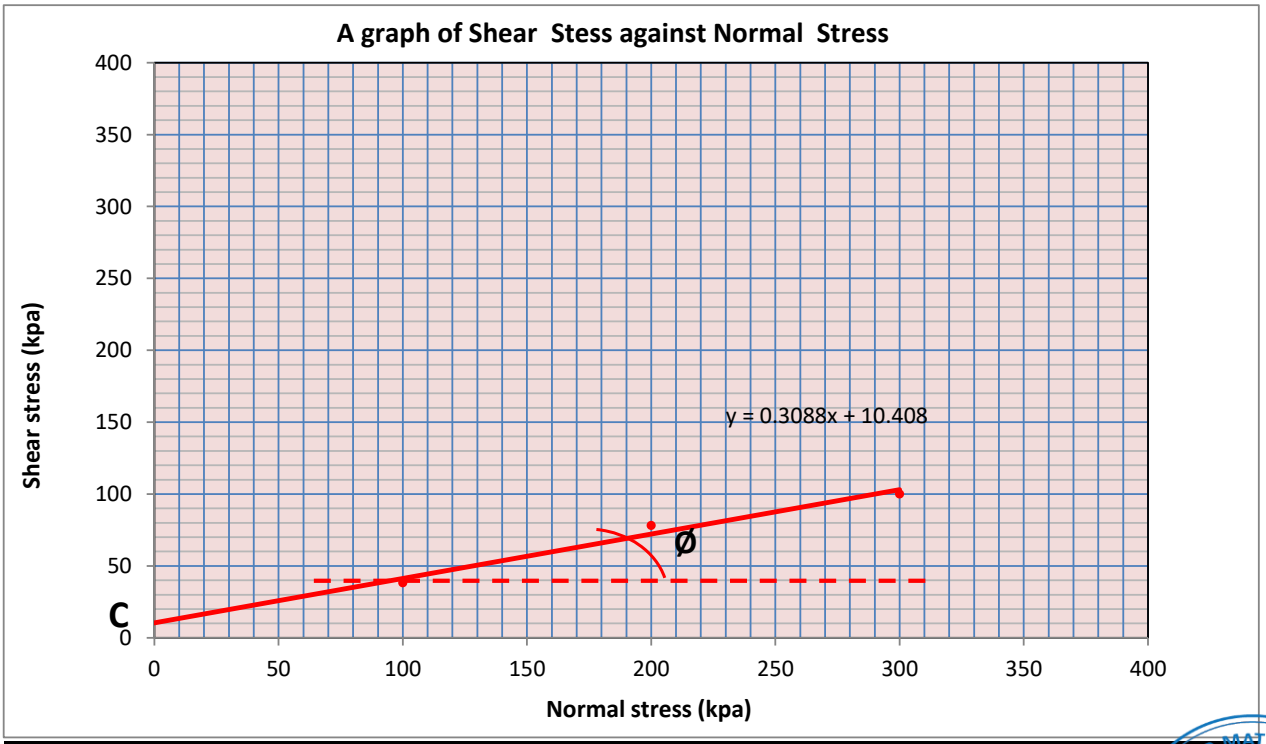
PROPOSED CONSTRUCTION OF KIRIAINI AHP IN MATHIOYA CONSTITUENCY, MURANG'A COUNTY WITH ASSOCIATED AMENITIES AND INFRASTRUCTURE

Sampled and submitted by:	CRANSFIELD	**Date submitted:**	10-Jun-26	**Date tested:**	12-Jun-26
Sample ID:	BH 01	**Sample depth (M):**	1.5	**Sample No:**	_

SPECIMEN AND TEST DETAILS

Sample Length; L	**(cm):**	6.00	**Natural Moisture Content**	**(%):**	37.6
Sample height	**(cm):**	2.50	**Bulk Density**	**(g/m^3):**	1823
Sample Area; A_o	**(cm^2):**	36.00	**Maximum Dry Density**	**(g/m^3):**	1325
Sample Volume	**(cm^3):**	90.00	**Mass of specimen in shear ring**	**(g):**	164.09
Shear Rate	**(mm/min):**		**Specific gravity - Assumed**	**(g/m^3):**	2.60

Max. load cell Reading (KN)	0.121 0.247 0.314	**Specimen Displacement (cm)**	0.733 0.733 0.769	**Corrected Specimen Area (cm^2)**	31.60 31.60 31.39	**Normal Stress (kpa):**	100 200 300	**Shear stress (kpa):**	38.29 78.16 100.04



C	10 **kpa**
Ø	17 °
SBP	**121 KN/m2**

Tested by:	PAUL	**Date Reported:**	14-Jun-26	**Checked by:**

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Email: c.materialcenter@gmail.com
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UNCONSOLIATED DIRECT SHEAR TEST (BS1377-7:1990,4)

STATE DEPARTMENT FOR HOUSING AND URBAN DEVELOPMENT

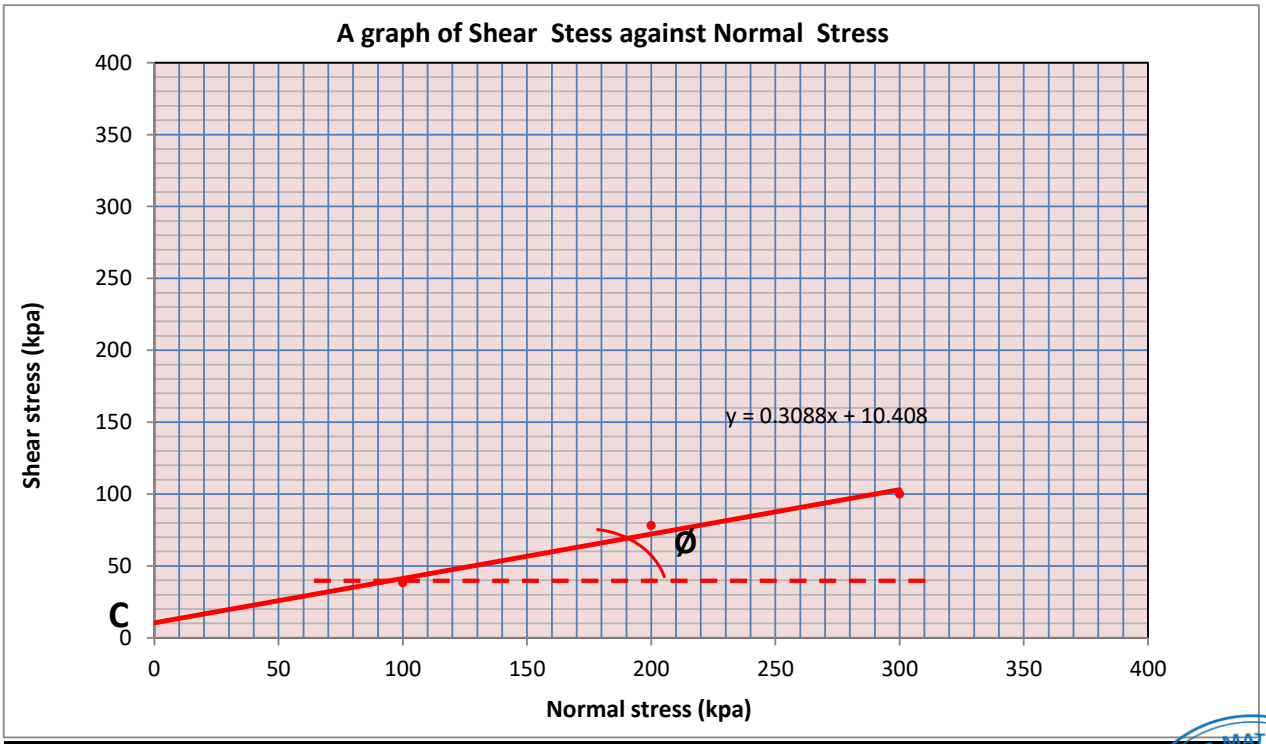
PROPOSED CONSTRUCTION OF KIRIAINI AHP IN MATHIOYA CONSTITUENCY, MURANG'A COUNTY WITH ASSOCIATED AMENITIES AND INFRASTRUCTURE

Sampled and submitted by:	CRANSFIELD	**Date submitted:**	10-Jun-26	**Date tested:**	12-Jun-26
Sample ID:	BH 01	**Sample depth (M):**	1.5	**Sample No:**	_

SPECIMEN AND TEST DETAILS

Sample Length; L (cm):	6.00	**Natural Moisture Content (%):**	37.6
Sample height (cm):	2.50	**Bulk Density (g/m^3):**	1823
Sample Area; A_o (cm^2):	36.00	**Maximum Dry Density (g/m^3):**	1325
Sample Volume (cm^3):	90.00	**Mass of specimen in shear ring (g):**	164.09
Shear Rate (mm/min):		**Specific gravity - Assumed (g/m^3):**	2.60

Max. load cell Reading (KN)	0.121 0.247 0.314	**Specimen Displacement (cm)**	0.733 0.733 0.769	**Corrected Specimen Area (cm^2)**	31.60 31.60 31.39	**Normal Stress (kpa):**	100 200 300	**Shear stress (kpa):**	38.29 78.16 100.04



C	10 **kpa**
Ø	17 °
SBP	**126 KN/m2**

Tested by:	PAUL	**Date Reported:**	14-Jun-26	**Checked by:** _____

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Sign:.....
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UNCONSOLIATED DIRECT SHEAR TEST (BS1377-7:1990,4)

STATE DEPARTMENT FOR HOUSING AND URBAN DEVELOPMENT

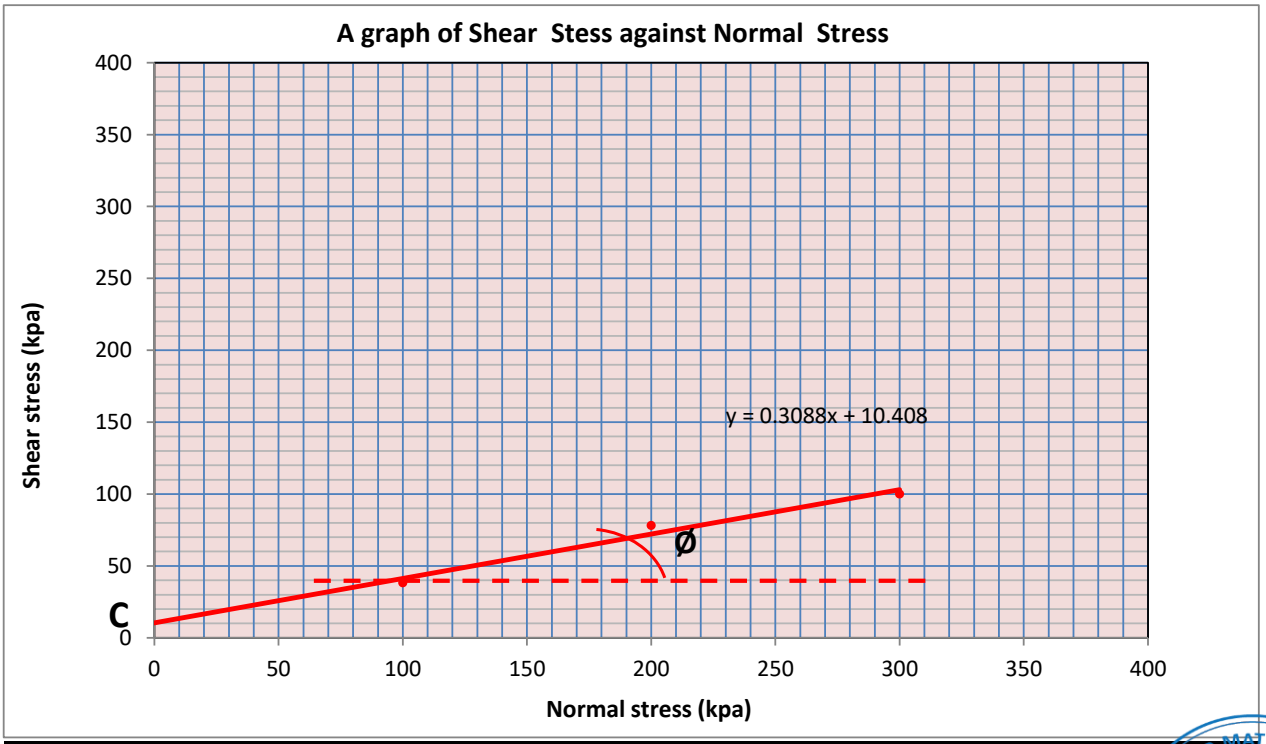
PROPOSED CONSTRUCTION OF KIRIAINI AHP IN MATHIOYA CONSTITUENCY, MURANG'A COUNTY WITH ASSOCIATED AMENITIES AND INFRASTRUCTURE

Sampled and submitted by:	CRANSFIELD	**Date submitted:**	10-Jun-26	**Date tested:**	12-Jun-26
Sample ID:	BH 01	**Sample depth (M):**	1.5	**Sample No:**	_

SPECIMEN AND TEST DETAILS

Sample Length; L (cm):	6.00	**Natural Moisture Content (%):**	37.6
Sample height (cm):	2.50	**Bulk Density (g/m^3):**	1823
Sample Area; A_o (cm^2):	36.00	**Maximum Dry Density (g/m^3):**	1325
Sample Volume (cm^3):	90.00	**Mass of specimen in shear ring (g):**	164.09
Shear Rate (mm/min):		**Specific gravity - Assumed (g/m^3):**	2.60

Max. load cell Reading (KN)	0.121 0.247 0.314	**Specimen Displacement (cm)**	0.733 0.733 0.769	**Corrected Specimen Area (cm^2)**	31.60 31.60 31.39	**Normal Stress (kpa):**	100 200 300	**Shear stress (kpa):**	38.29 78.16 100.04



C	10 **kpa**
Ø	17 °
SBP	**131 KN/m2**

Tested by:	PAUL	**Date Reported:**	14-Jun-26	**Checked by:**

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Sign:
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UNCONSOLIATED DIRECT SHEAR TEST (BS1377-7:1990,4)

STATE DEPARTMENT FOR HOUSING AND URBAN DEVELOPMENT

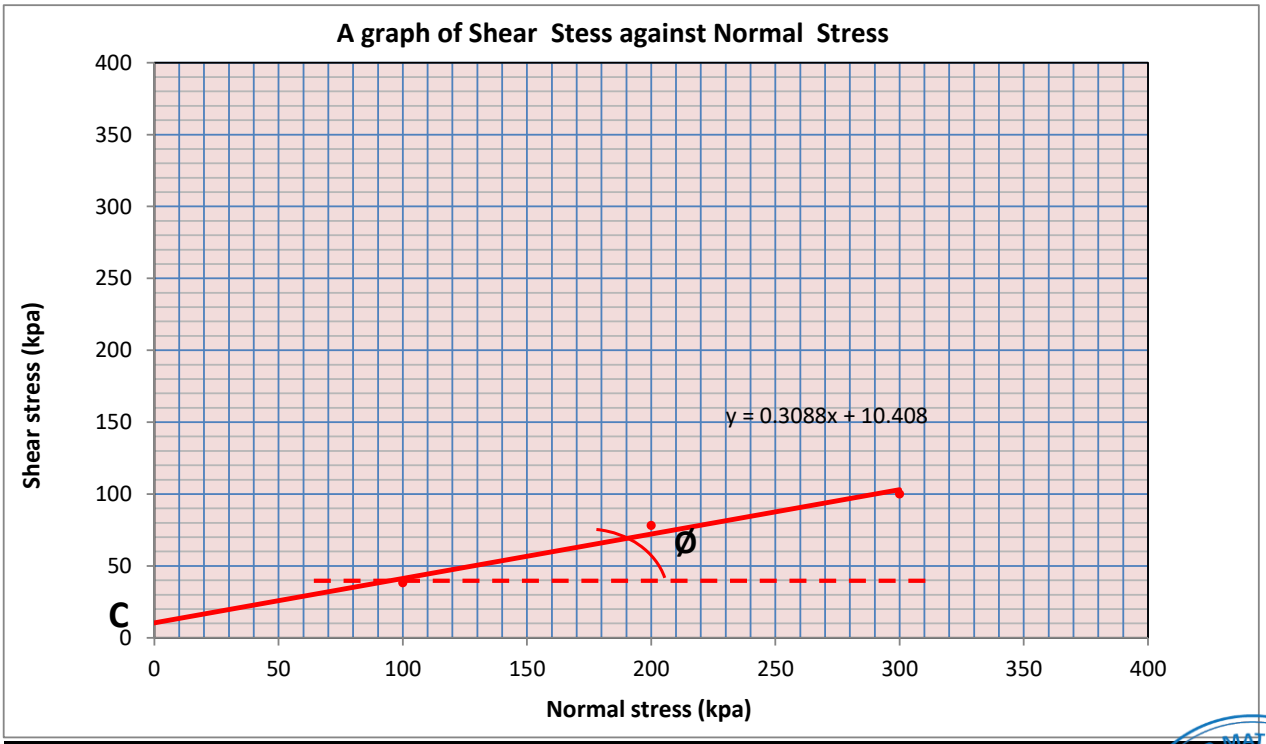
PROPOSED CONSTRUCTION OF KIRIAINI AHP IN MATHIOYA CONSTITUENCY, MURANG'A COUNTY WITH ASSOCIATED AMENITIES AND INFRASTRUCTURE

Sampled and submitted by:	CRANSFIELD	**Date submitted:**	10-Jun-26	**Date tested:**	12-Jun-26
Sample ID:	BH 01	**Sample depth (M):**	2.0	**Sample No:**	_

SPECIMEN AND TEST DETAILS

Sample Length; L (cm):	6.00	**Natural Moisture Content (%):**	37.6
Sample height (cm):	2.50	**Bulk Density (g/m^3):**	1823
Sample Area; A_o (cm^2):	36.00	**Maximum Dry Density (g/m^3):**	1325
Sample Volume (cm^3):	90.00	**Mass of specimen in shear ring (g):**	164.09
Shear Rate (mm/min):		**Specific gravity - Assumed (g/m^3):**	2.60

Max. load cell Reading (KN)	0.121 0.247 0.314	**Specimen Displacement (cm)**	0.733 0.733 0.769	**Corrected Specimen Area (cm^2)**	31.60 31.60 31.39	**Normal Stress (kpa):**	100 200 300	**Shear stress (kpa):**	38.29 78.16 100.04



C	10 **kpa**
Ø	17 °
SBP	**132 KN/m2**

Tested by:	PAUL	**Date Reported:**	14-Jun-26	**Checked by:**

CRANSFIELD MATERIAL TESTING
Email: c.materialcenter@gmail.com
Sign:
P.O. Box 59401-00100 NAIROBI

UNCONSOLIATED DIRECT SHEAR TEST (BS1377-7:1990,4)

STATE DEPARTMENT FOR HOUSING AND URBAN DEVELOPMENT

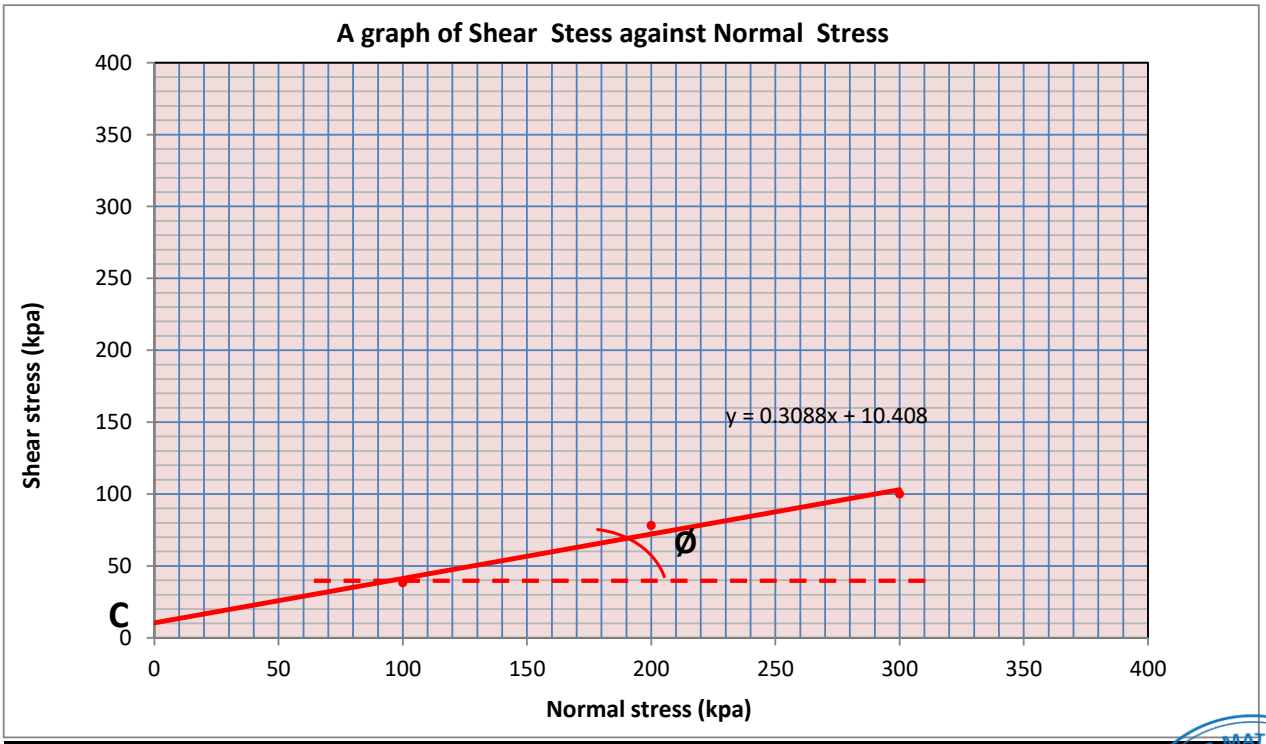
PROPOSED CONSTRUCTION OF KIRIAINI AHP IN MATHIOYA CONSTITUENCY, MURANG'A COUNTY WITH ASSOCIATED AMENITIES AND INFRASTRUCTURE

Sampled and submitted by:	CRANSFIELD	**Date submitted:**	10-Jun-26	**Date tested:**	12-Jun-26
Sample ID:	BH 01	**Sample depth (M):**	2.0	**Sample No:**	_

SPECIMEN AND TEST DETAILS

Sample Length; L (cm):	6.00	**Natural Moisture Content (%):**	37.6
Sample height (cm):	2.50	**Bulk Density (g/m^3):**	1823
Sample Area; A_o (cm^2):	36.00	**Maximum Dry Density (g/m^3):**	1325
Sample Volume (cm^3):	90.00	**Mass of specimen in shear ring (g):**	164.09
Shear Rate (mm/min):		**Specific gravity - Assumed (g/m^3):**	2.60

Max. load cell Reading (KN)	0.121 0.247 0.314	**Specimen Displacement (cm)**	0.733 0.733 0.769	**Corrected Specimen Area (cm^2)**	31.60 31.60 31.39	**Normal Stress (kpa):**	100 200 300	**Shear stress (kpa):**	38.29 78.16 100.04



C	10 **kpa**
Ø	17 °
SBP	**137 KN/m2**

Tested by:	PAUL	**Date Reported:**	14-Jun-26	**Checked by:**

CRANSFIELD MATERIAL TESTING
Email: c.materialcenter@gmail.com
Sign:
P.O. Box 59401-00100 NAIROBI

UNCONSOLIATED DIRECT SHEAR TEST (BS1377-7:1990,4)

STATE DEPARTMENT FOR HOUSING AND URBAN DEVELOPMENT

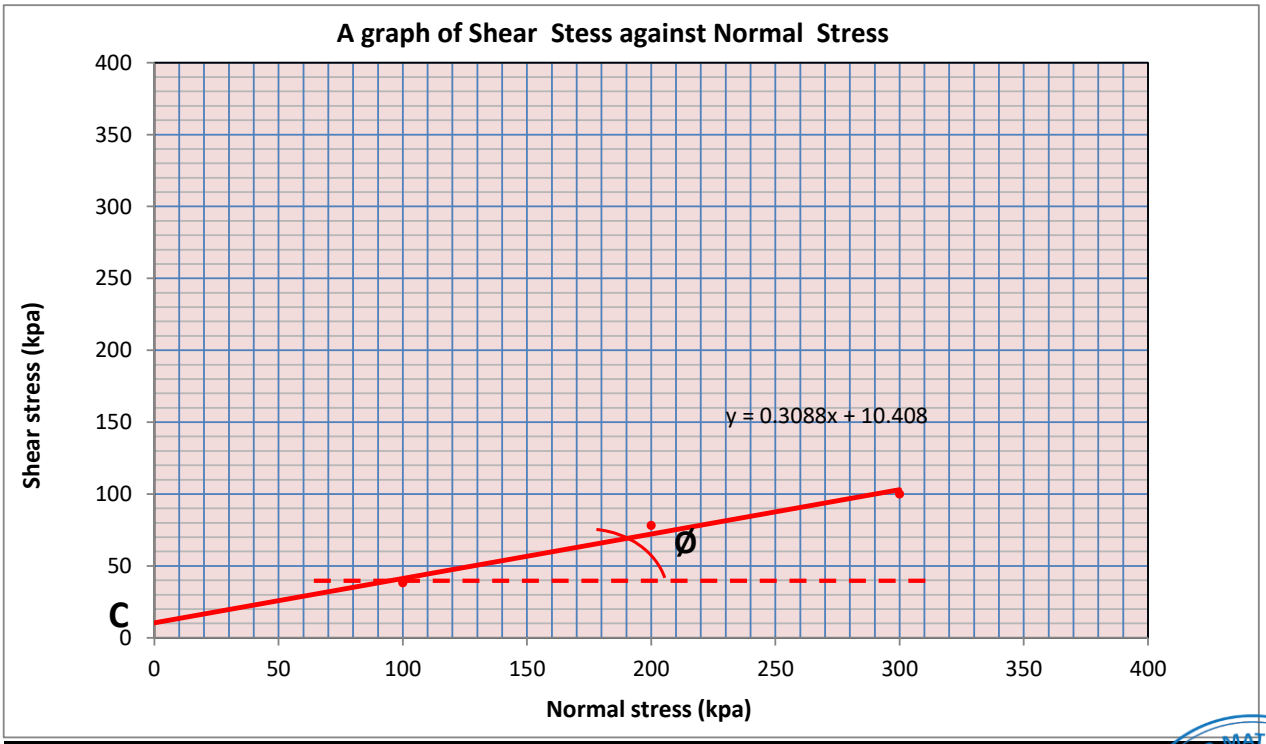
PROPOSED CONSTRUCTION OF KIRIAINI AHP IN MATHIOYA CONSTITUENCY, MURANG'A COUNTY WITH ASSOCIATED AMENITIES AND INFRASTRUCTURE

Sampled and submitted by:	CRANSFIELD	**Date submitted:**	10-Jun-26	**Date tested:**	12-Jun-26
Sample ID:	BH 01	**Sample depth (M):**	2.0	**Sample No:**	_

SPECIMEN AND TEST DETAILS

Sample Length; L (cm):	6.00	**Natural Moisture Content (%):**	37.6
Sample height (cm):	2.50	**Bulk Density (g/m^3):**	1823
Sample Area; A_o (cm^2):	36.00	**Maximum Dry Density (g/m^3):**	1325
Sample Volume (cm^3):	90.00	**Mass of specimen in shear ring (g):**	164.09
Shear Rate (mm/min):		**Specific gravity - Assumed (g/m^3):**	2.60

Max. load cell Reading (KN)	0.121 0.247 0.314	**Specimen Displacement (cm)**	0.733 0.733 0.769	**Corrected Specimen Area (cm^2)**	31.60 31.60 31.39	**Normal Stress (kpa):**	100 200 300	**Shear stress (kpa):**	38.29 78.16 100.04



C	10 **kpa**
Ø	17 °
SBP	**142** **KN/m2**

Tested by:	PAUL	**Date Reported:**	14-Jun-26	**Checked by:**

CRANSFIELD MATERIAL TESTING
Email: c.materialcenter@gmail.com
Sign:
P.O. Box 59401-00100 NAIROBI

UNCONSOLIATED DIRECT SHEAR TEST (BS1377-7:1990,4)

STATE DEPARTMENT FOR HOUSING AND URBAN DEVELOPMENT

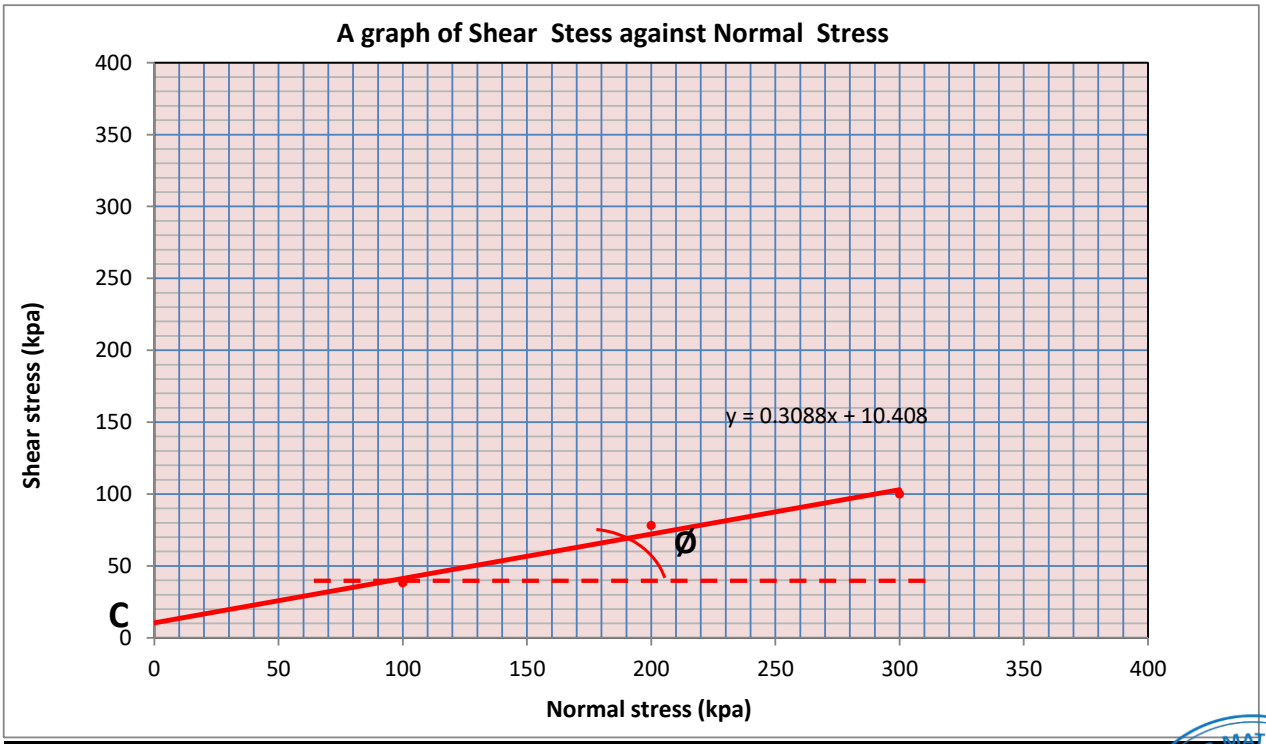
PROPOSED CONSTRUCTION OF KIRIAINI AHP IN MATHIOYA CONSTITUENCY, MURANG'A COUNTY WITH ASSOCIATED AMENITIES AND INFRASTRUCTURE

Sampled and submitted by:	CRANSFIELD	**Date submitted:**	10-Jun-26	**Date tested:**	12-Jun-26
Sample ID:	BH 01	**Sample depth (M):**	2.0	**Sample No:**	_

SPECIMEN AND TEST DETAILS

Sample Length; L (cm):	6.00	**Natural Moisture Content (%):**	37.6
Sample height (cm):	2.50	**Bulk Density (g/m^3):**	1823
Sample Area; A_O (cm^2):	36.00	**Maximum Dry Density (g/m^3):**	1325
Sample Volume (cm^3):	90.00	**Mass of specimen in shear ring (g):**	164.09
Shear Rate (mm/min):		**Specific gravity - Assumed (g/m^3):**	2.60

Max. load cell Reading (KN)	0.121 0.247 0.314	**Specimen Displacement (cm)**	0.733 0.733 0.769	**Corrected Specimen Area (cm^2)**	31.60 31.60 31.39	**Normal Stress (kpa):**	100 200 300	**Shear stress (kpa):**	38.29 78.16 100.04



C	10 **kpa**
Ø	17 °
SBP	**148 KN/m2**

Tested by:	PAUL	**Date Reported:**	14-Jun-26	**Checked by:**

CRANSFIELD MATERIAL TESTING
Email: c.materialcenter@gmail.com
Sign:
P.O. Box 59401-00100 NAIROBI

UNCONSOLIATED DIRECT SHEAR TEST (BS1377-7:1990,4)

STATE DEPARTMENT FOR HOUSING AND URBAN DEVELOPMENT

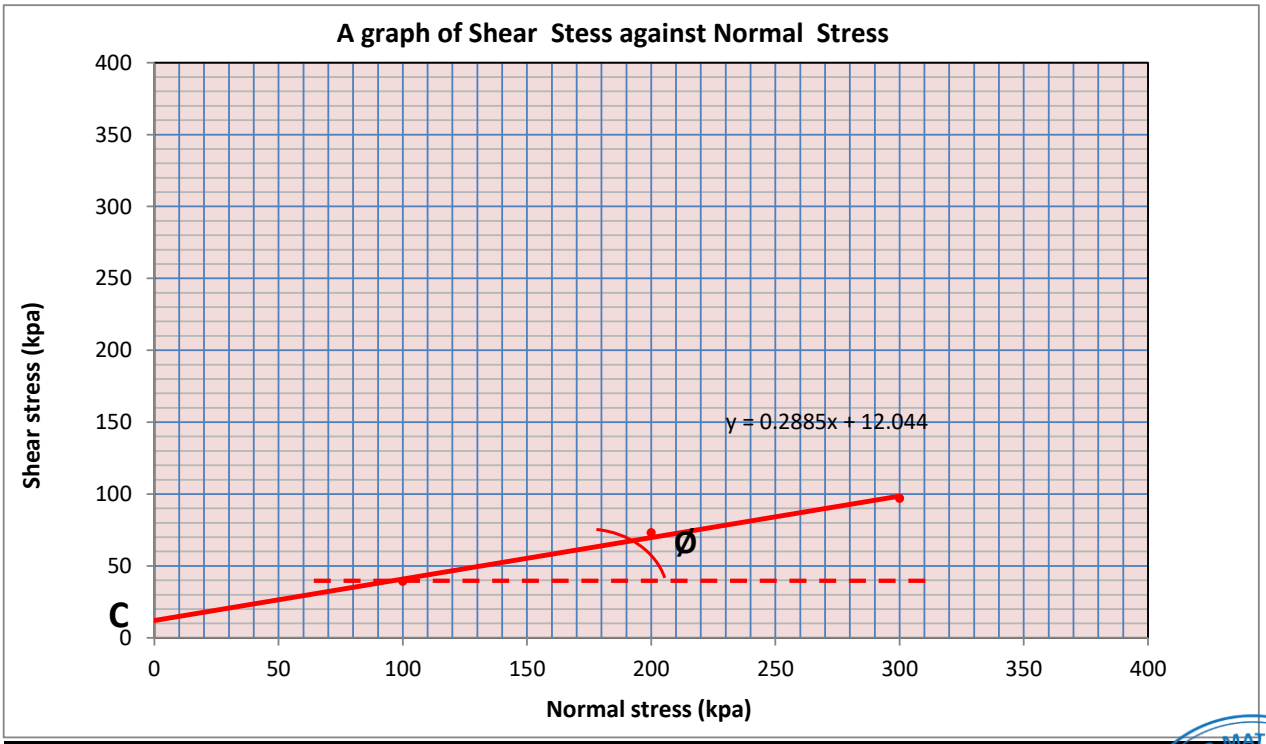
PROPOSED CONSTRUCTION OF KIRIAINI AHP IN MATHIOYA CONSTITUENCY, MURANG'A COUNTY WITH ASSOCIATED AMENITIES AND INFRASTRUCTURE

Sampled and submitted by:	CRANSFIELD	**Date submitted:**	10-Jun-26	**Date tested:**	12-Jun-26
Sample ID:	BH 02	**Sample depth (M):**	1.5	**Sample No:**	_

SPECIMEN AND TEST DETAILS

Sample Length; L (cm):	6.00	**Natural Moisture Content (%):**	27.3
Sample height (cm):	2.50	**Bulk Density (g/m^3):**	1853
Sample Area; A_o (cm^2):	36.00	**Maximum Dry Density (g/m^3):**	1456
Sample Volume (cm^3):	90.00	**Mass of specimen in shear ring (g):**	166.81
Shear Rate (mm/min):		**Specific gravity - Assumed (g/m^3):**	2.60

Max. load cell Reading (KN)	0.124 0.231 0.304	**Specimen Displacement (cm)**	0.731 0.731 0.772	**Corrected Specimen Area (cm^2)**	31.61 31.61 31.37	**Normal Stress (kpa):**	100 200 300	**Shear stress (kpa):**	39.22 73.07 96.91



C	12 **kpa**
Ø	16 °
SBP	**119 KN/m2**

Tested by:	PAUL	**Date Reported:**	14-Jun-26	**Checked by:**

CRANSFIELD MATERIAL TESTING
Email: c.materialcenter@gmail.com
Sign:
P.O. Box 59401-00100 NAIROBI

UNCONSOLIATED DIRECT SHEAR TEST (BS1377-7:1990,4)

STATE DEPARTMENT FOR HOUSING AND URBAN DEVELOPMENT

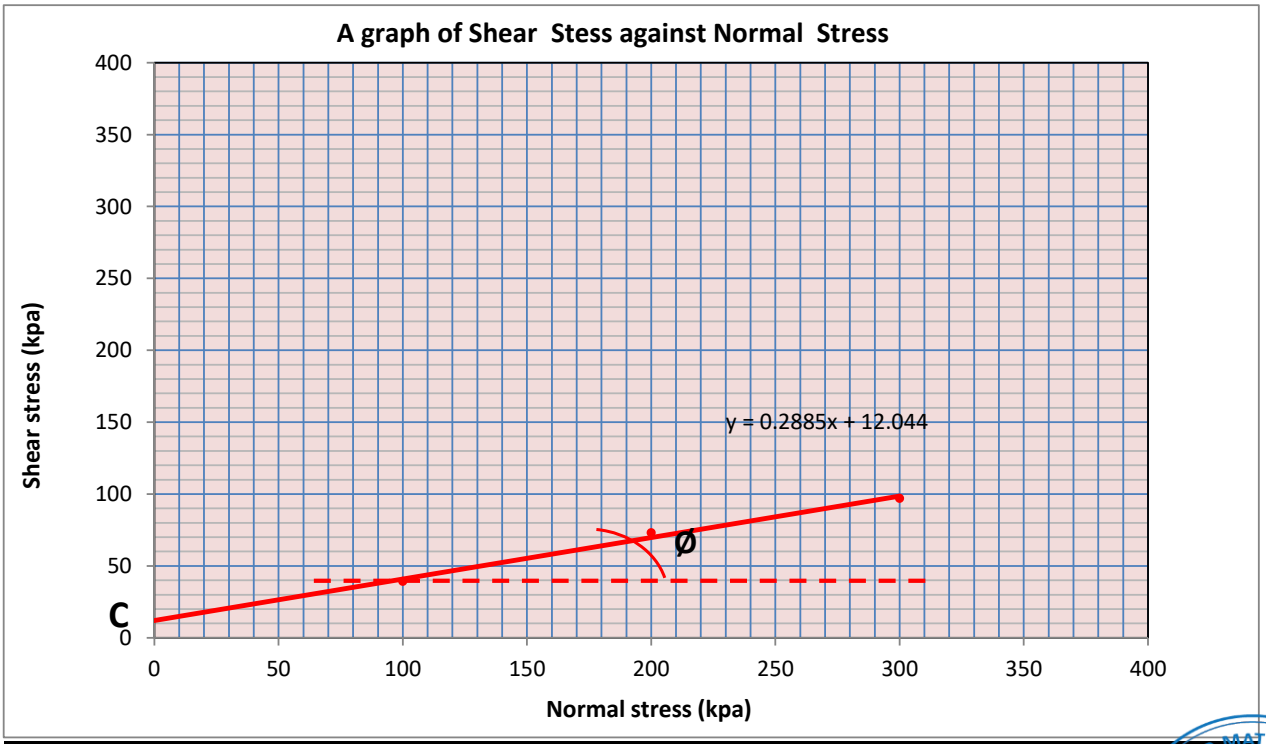
PROPOSED CONSTRUCTION OF KIRIAINI AHP IN MATHIOYA CONSTITUENCY, MURANG'A COUNTY WITH ASSOCIATED AMENITIES AND INFRASTRUCTURE

Sampled and submitted by:	CRANSFIELD	**Date submitted:**	10-Jun-26	**Date tested:**	12-Jun-26
Sample ID:	BH 02	**Sample depth (M):**	1.5	**Sample No:**	_

SPECIMEN AND TEST DETAILS

Sample Length; L	**(cm):**	6.00	**Natural Moisture Content**	**(%):**	27.3
Sample height	**(cm):**	2.50	**Bulk Density**	**(g/m^3):**	1853
Sample Area; A_O	**(cm^2):**	36.00	**Maximum Dry Density**	**(g/m^3):**	1456
Sample Volume	**(cm^3):**	90.00	**Mass of specimen in shear ring**	**(g):**	166.81
Shear Rate	**(mm/min):**		**Specific gravity - Assumed**	**(g/m^3):**	2.60

Max. load cell Reading (KN)	0.124 0.231 0.304	**Specimen Displacement (cm)**	0.731 0.731 0.772	**Corrected Specimen Area (cm^2)**	31.61 31.61 31.37	**Normal Stress (kpa):**	100 200 300	**Shear stress (kpa):**	39.22 73.07 96.91



C	12 **kpa**
Ø	16 °
SBP	**123 KN/m2**

Tested by:	PAUL	**Date Reported:**	14-Jun-26	**Checked by:**

CRANSFIELD MATERIAL TESTING
Email: c.materialcenter@gmail.com
Sign:
P.O. Box 59401-00100 NAIROBI

UNCONSOLIATED DIRECT SHEAR TEST (BS1377-7:1990,4)

STATE DEPARTMENT FOR HOUSING AND URBAN DEVELOPMENT

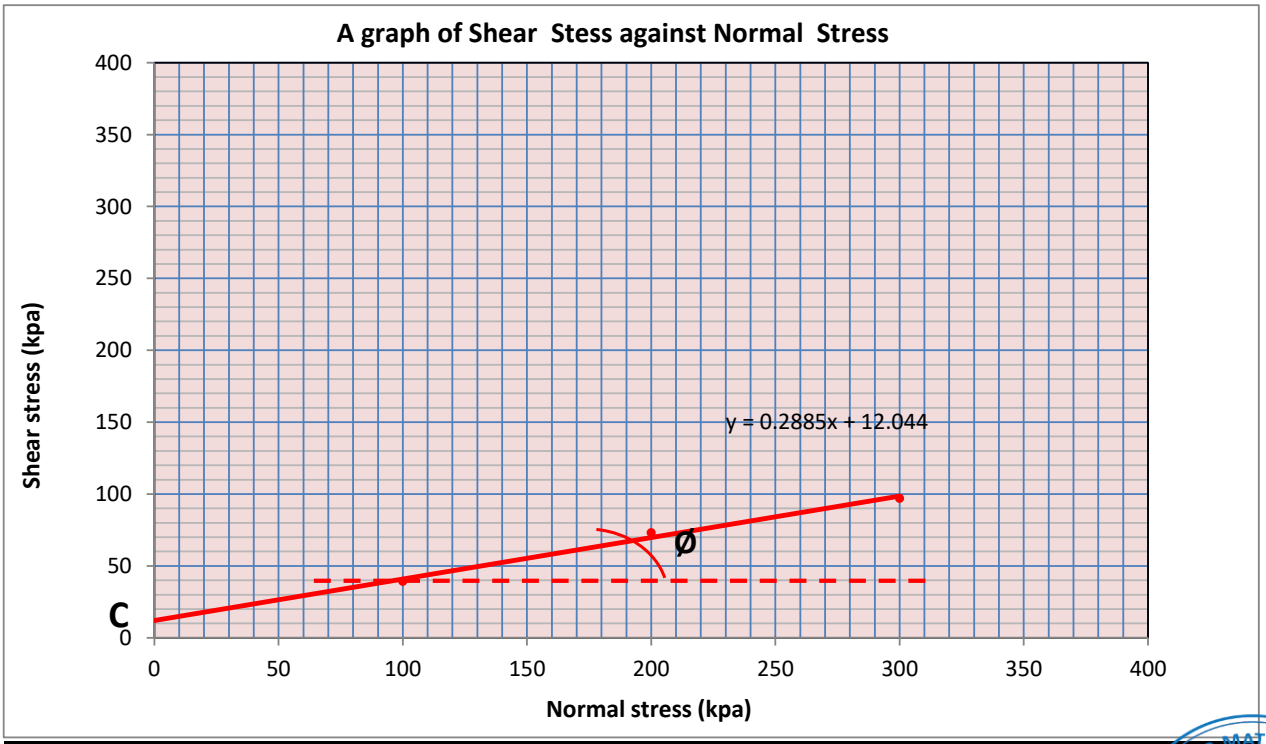
PROPOSED CONSTRUCTION OF KIRIAINI AHP IN MATHIOYA CONSTITUENCY, MURANG'A COUNTY WITH ASSOCIATED AMENITIES AND INFRASTRUCTURE

Sampled and submitted by:	CRANSFIELD	**Date submitted:**	10-Jun-26	**Date tested:**	12-Jun-26
Sample ID:	BH 02	**Sample depth (M):**	1.5	**Sample No:**	_

SPECIMEN AND TEST DETAILS

Sample Length; L	**(cm):**	6.00	**Natural Moisture Content**	**(%):**	27.3
Sample height	**(cm):**	2.50	**Bulk Density**	**(g/m^3):**	1853
Sample Area; A_o	**(cm^2):**	36.00	**Maximum Dry Density**	**(g/m^3):**	1456
Sample Volume	**(cm^3):**	90.00	**Mass of specimen in shear ring**	**(g):**	166.81
Shear Rate	**(mm/min):**		**Specific gravity - Assumed**	**(g/m^3):**	2.60

Max. load cell Reading (KN)	0.124 0.231 0.304	**Specimen Displacement (cm)**	0.731 0.731 0.772	**Corrected Specimen Area (cm^2)**	31.61 31.61 31.37	**Normal Stress (kpa):**	100 200 300	**Shear stress (kpa):**	39.22 73.07 96.91



C	12 **kpa**
Ø	16 °
SBP	**128 KN/m2**

Tested by:	PAUL	**Date Reported:**	14-Jun-26	**Checked by:**

CRANSFIELD MATERIAL TESTING
Email: c.materialcenter@gmail.com
Sign:
P.O. Box 59401-00100 NAIROBI

UNCONSOLIATED DIRECT SHEAR TEST (BS1377-7:1990,4)

STATE DEPARTMENT FOR HOUSING AND URBAN DEVELOPMENT

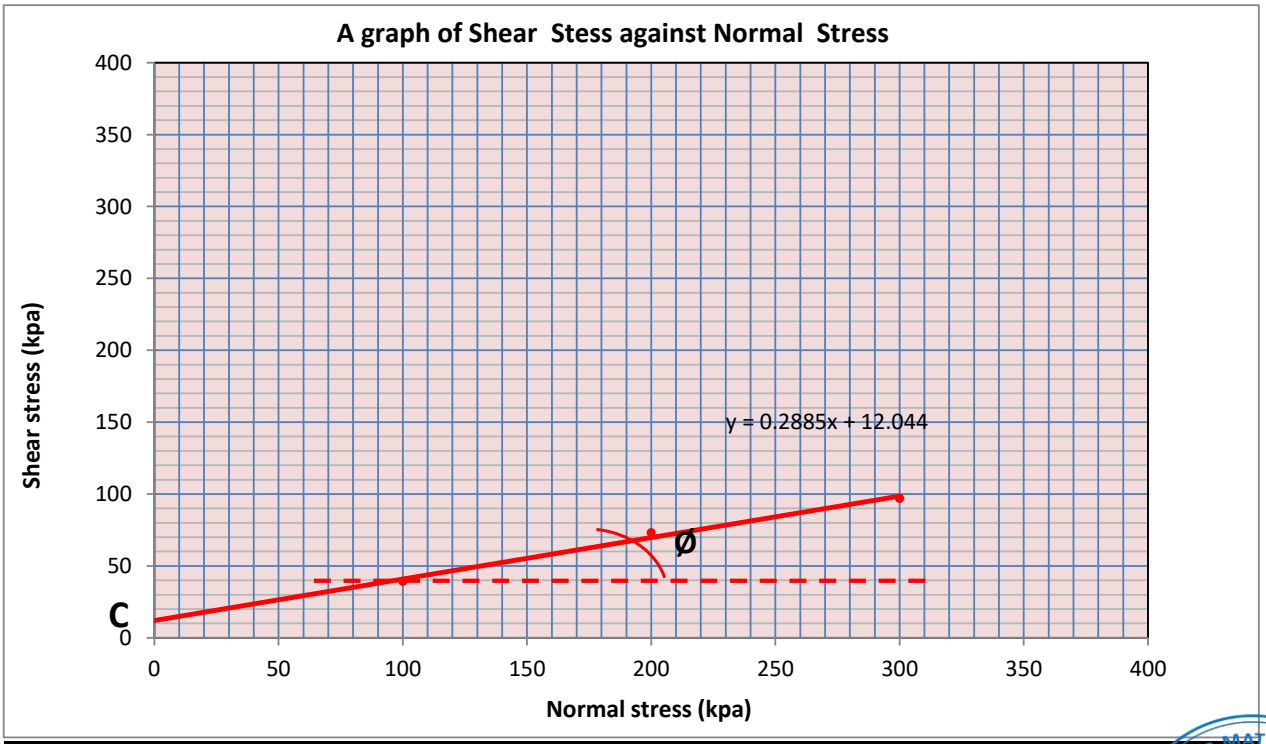
PROPOSED CONSTRUCTION OF KIRIAINI AHP IN MATHIOYA CONSTITUENCY, MURANG'A COUNTY WITH ASSOCIATED AMENITIES AND INFRASTRUCTURE

Sampled and submitted by:	CRANSFIELD	**Date submitted:**	10-Jun-26	**Date tested:**	12-Jun-26
Sample ID:	BH 02	**Sample depth (M):**	1.5	**Sample No:**	_

SPECIMEN AND TEST DETAILS

Sample Length; L	**(cm):**	6.00	**Natural Moisture Content**	**(%):**	27.3
Sample height	**(cm):**	2.50	**Bulk Density**	**(g/m^3):**	1853
Sample Area; A_o	**(cm^2):**	36.00	**Maximum Dry Density**	**(g/m^3):**	1456
Sample Volume	**(cm^3):**	90.00	**Mass of specimen in shear ring**	**(g):**	166.81
Shear Rate	**(mm/min):**		**Specific gravity - Assumed**	**(g/m^3):**	2.60

Max. load cell Reading (KN)	0.124 0.231 0.304	**Specimen Displacement (cm)**	0.731 0.731 0.772	**Corrected Specimen Area (cm^2)**	31.61 31.61 31.37	**Normal Stress (kpa):**	100 200 300	**Shear stress (kpa):**	39.22 73.07 96.91



C	12 **kpa**
Ø	16 °
SBP	**132 KN/m2**

Tested by:	PAUL	**Date Reported:**	14-Jun-26	**Checked by:**

CRANSFIELD MATERIAL TESTING
Email: c.materialcenter@gmail.com
Sign:
P.O. Box 59401-00100 NAIROBI

UNCONSOLIATED DIRECT SHEAR TEST (BS1377-7:1990,4)

STATE DEPARTMENT FOR HOUSING AND URBAN DEVELOPMENT

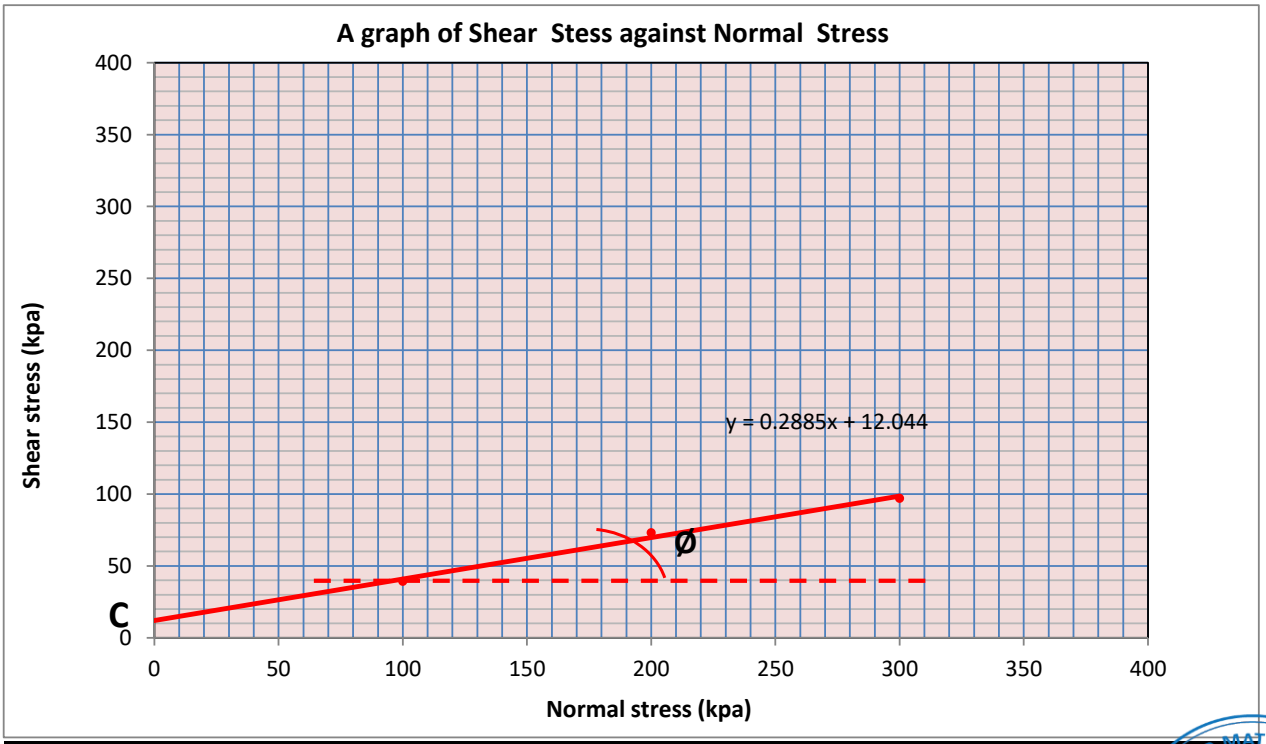
PROPOSED CONSTRUCTION OF KIRIAINI AHP IN MATHIOYA CONSTITUENCY, MURANG'A COUNTY WITH ASSOCIATED AMENITIES AND INFRASTRUCTURE

Sampled and submitted by:	CRANSFIELD	**Date submitted:**	10-Jun-26	**Date tested:**	12-Jun-26
Sample ID:	BH 02	**Sample depth (M):**	2.0	**Sample No:**	_

SPECIMEN AND TEST DETAILS

Sample Length; L	**(cm):**	6.00	**Natural Moisture Content**	**(%):**	27.3
Sample height	**(cm):**	2.50	**Bulk Density**	**(g/m^3):**	1853
Sample Area; A_O	**(cm^2):**	36.00	**Maximum Dry Density**	**(g/m^3):**	1456
Sample Volume	**(cm^3):**	90.00	**Mass of specimen in shear ring**	**(g):**	166.81
Shear Rate	**(mm/min):**		**Specific gravity - Assumed**	**(g/m^3):**	2.60

Max. load cell Reading (KN)	0.124 0.231 0.304	**Specimen Displacement (cm)**	0.731 0.731 0.772	**Corrected Specimen Area (cm^2)**	31.61 31.61 31.37	**Normal Stress (kpa):**	100 200 300	**Shear stress (kpa):**	39.22 73.07 96.91



C	12 **kpa**
Ø	16 °
SBP	**134 KN/m2**

Tested by:	PAUL	**Date Reported:**	14-Jun-26	**Checked by:**

CRANSFIELD MATERIAL TESTING
Email: c.materialcenter@gmail.com
Sign:
P.O. Box 59401-00100 NAIROBI

UNCONSOLIATED DIRECT SHEAR TEST (BS1377-7:1990,4)

STATE DEPARTMENT FOR HOUSING AND URBAN DEVELOPMENT

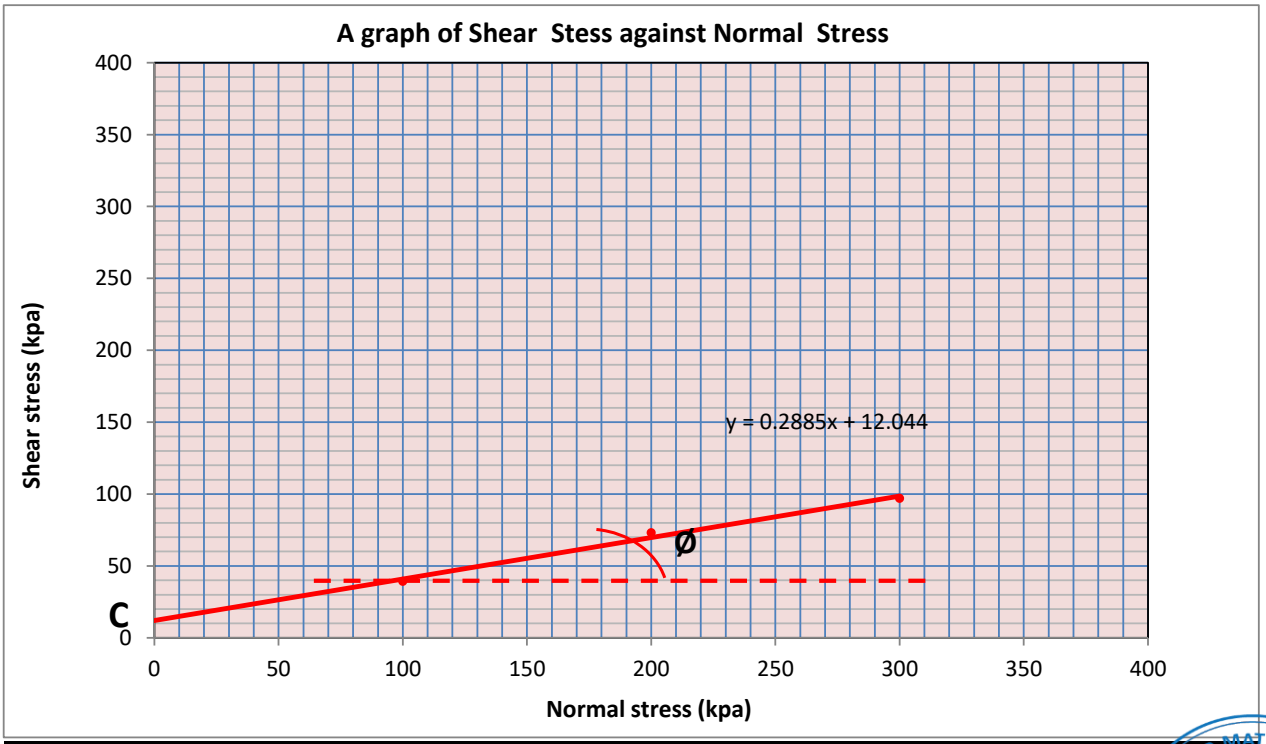
PROPOSED CONSTRUCTION OF KIRIAINI AHP IN MATHIOYA CONSTITUENCY, MURANG'A COUNTY WITH ASSOCIATED AMENITIES AND INFRASTRUCTURE

Sampled and submitted by:	CRANSFIELD	**Date submitted:**	10-Jun-26	**Date tested:**	12-Jun-26
Sample ID:	BH 02	**Sample depth (M):**	2.0	**Sample No:**	_

SPECIMEN AND TEST DETAILS

Sample Length; L	**(cm):**	6.00	**Natural Moisture Content**	**(%):**	27.3
Sample height	**(cm):**	2.50	**Bulk Density**	**(g/m^3):**	1853
Sample Area; A_O	**(cm^2):**	36.00	**Maximum Dry Density**	**(g/m^3):**	1456
Sample Volume	**(cm^3):**	90.00	**Mass of specimen in shear ring**	**(g):**	166.81
Shear Rate	**(mm/min):**		**Specific gravity - Assumed**	**(g/m^3):**	2.60

Max. load cell Reading (KN)	0.124 0.231 0.304	**Specimen Displacement (cm)**	0.731 0.731 0.772	**Corrected Specimen Area (cm^2)**	31.61 31.61 31.37	**Normal Stress (kpa):**	100 200 300	**Shear stress (kpa):**	39.22 73.07 96.91



C	12 **kpa**
Ø	16 °
SBP	**138 KN/m2**

Tested by:	PAUL	**Date Reported:**	14-Jun-26	**Checked by:**

CRANSFIELD MATERIAL TESTING
Email: c.materialcenter@gmail.com
Sign:
P.O. Box 59401-00100 NAIROBI

UNCONSOLIATED DIRECT SHEAR TEST (BS1377-7:1990,4)

STATE DEPARTMENT FOR HOUSING AND URBAN DEVELOPMENT

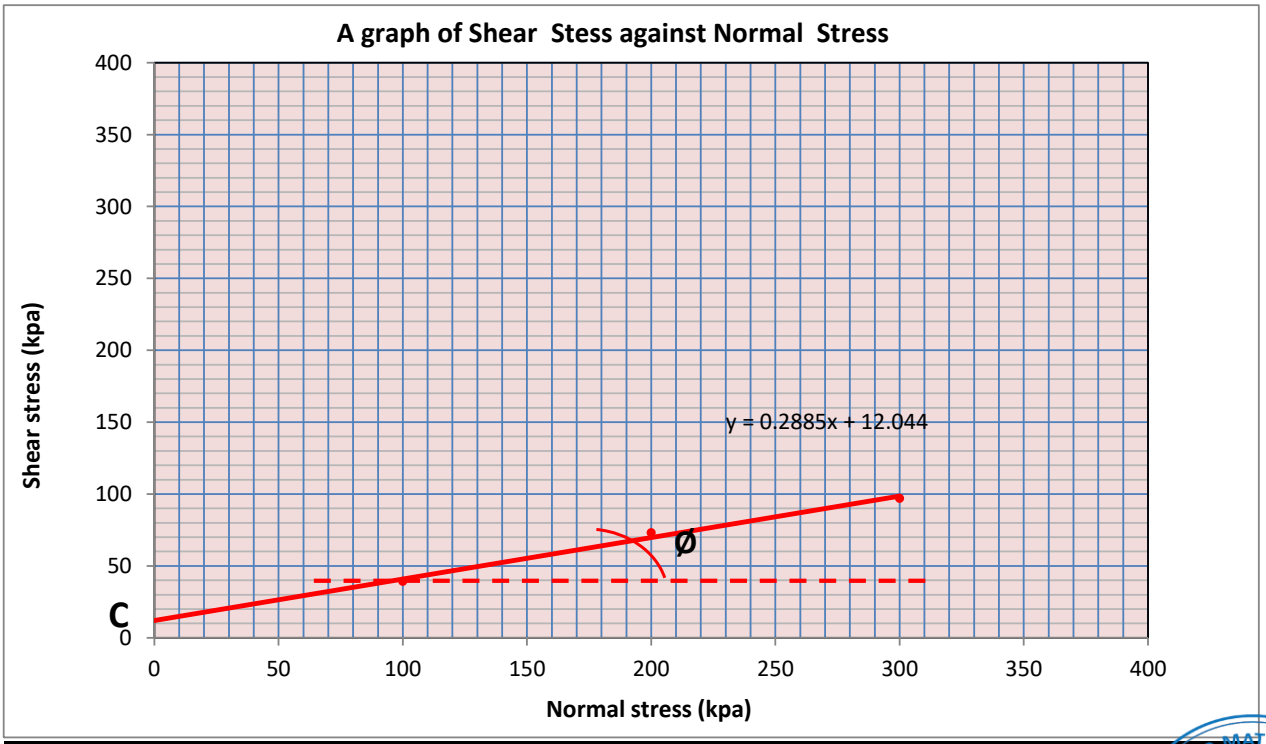
PROPOSED CONSTRUCTION OF KIRIAINI AHP IN MATHIOYA CONSTITUENCY, MURANG'A COUNTY WITH ASSOCIATED AMENITIES AND INFRASTRUCTURE

Sampled and submitted by:	CRANSFIELD	**Date submitted:**	10-Jun-26	**Date tested:**	12-Jun-26
Sample ID:	BH 02	**Sample depth (M):**	2.0	**Sample No:**	_

SPECIMEN AND TEST DETAILS

Sample Length; L	**(cm):**	6.00	**Natural Moisture Content**	**(%):**	27.3
Sample height	**(cm):**	2.50	**Bulk Density**	**(g/m^3):**	1853
Sample Area; A_O	**(cm^2):**	36.00	**Maximum Dry Density**	**(g/m^3):**	1456
Sample Volume	**(cm^3):**	90.00	**Mass of specimen in shear ring**	**(g):**	166.81
Shear Rate	**(mm/min):**		**Specific gravity - Assumed**	**(g/m^3):**	2.60

Max. load cell Reading (KN)	0.124 0.231 0.304	**Specimen Displacement (cm)**	0.731 0.731 0.772	**Corrected Specimen Area (cm^2)**	31.61 31.61 31.37	**Normal Stress (kpa):**	100 200 300	**Shear stress (kpa):**	39.22 73.07 96.91



C	12 **kpa**
Ø	16 °
SBP	**143 KN/m2**

Tested by:	PAUL	**Date Reported:**	14-Jun-26	**Checked by:**

CRANSFIELD MATERIAL TESTING
Email: c.materialcenter@gmail.com
Sign:
P.O. Box 59401-00100 NAIROBI

UNCONSOLIATED DIRECT SHEAR TEST (BS1377-7:1990,4)

STATE DEPARTMENT FOR HOUSING AND URBAN DEVELOPMENT

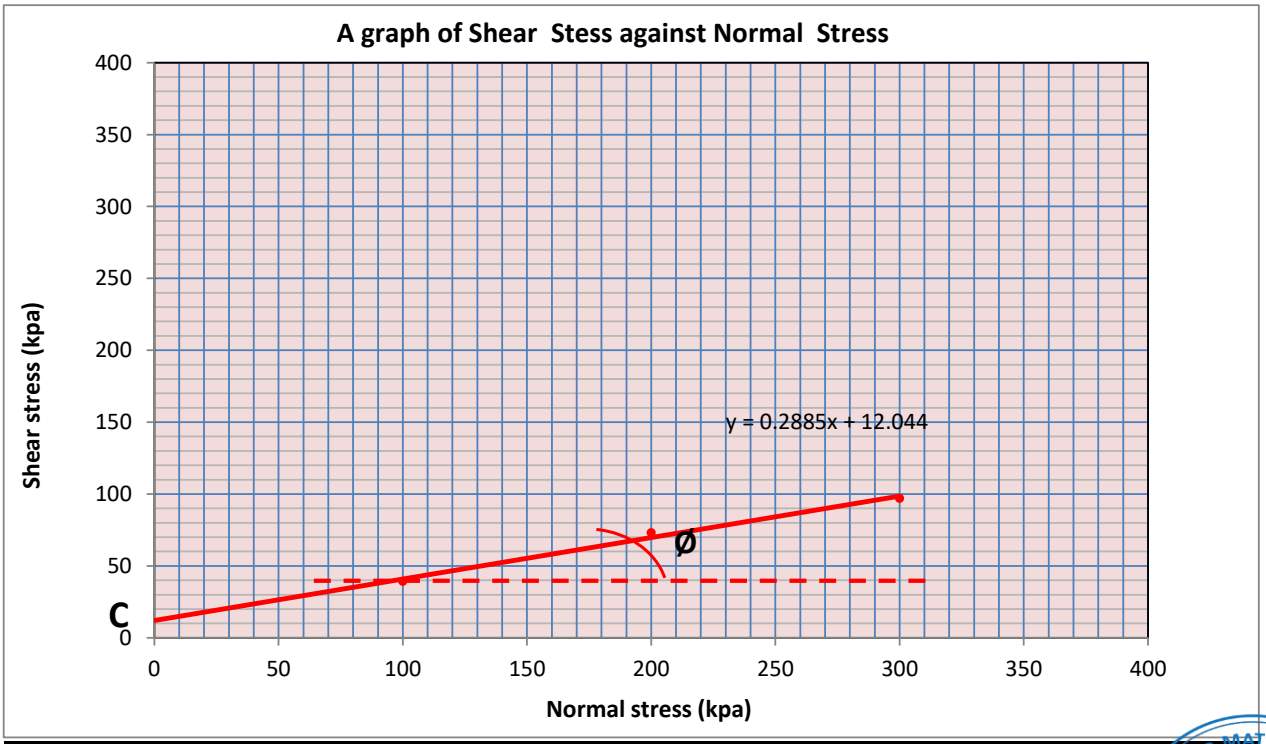
PROPOSED CONSTRUCTION OF KIRIAINI AHP IN MATHIOYA CONSTITUENCY, MURANG'A COUNTY WITH ASSOCIATED AMENITIES AND INFRASTRUCTURE

Sampled and submitted by:	CRANSFIELD	**Date submitted:**	10-Jun-26	**Date tested:**	12-Jun-26
Sample ID:	BH 02	**Sample depth (M):**	2.0	**Sample No:**	_

SPECIMEN AND TEST DETAILS

Sample Length; L	**(cm):**	6.00	**Natural Moisture Content**	**(%):**	27.3
Sample height	**(cm):**	2.50	**Bulk Density**	**(g/m^3):**	1853
Sample Area; A_O	**(cm^2):**	36.00	**Maximum Dry Density**	**(g/m^3):**	1456
Sample Volume	**(cm^3):**	90.00	**Mass of specimen in shear ring**	**(g):**	166.81
Shear Rate	**(mm/min):**		**Specific gravity - Assumed**	**(g/m^3):**	2.60

Max. load cell Reading (KN)	0.124 0.231 0.304	**Specimen Displacement (cm)**	0.731 0.731 0.772	**Corrected Specimen Area (cm^2)**	31.61 31.61 31.37	**Normal Stress (kpa):**	100 200 300	**Shear stress (kpa):**	39.22 73.07 96.91



C	12 **kpa**
Ø	16 °
SBP	**147 KN/m2**

Tested by:	PAUL	**Date Reported:**	14-Jun-26	**Checked by:**

CRANSFIELD MATERIAL TESTING
Email: c.materialcenter@gmail.com
Sign:
P.O. Box 59401-00100 NAIROBI

11. APPENDIX II (BOREHOLE LOGS)

Cransfield Materials Testing Center

CRANSFIELD CENTER
P.O Box 59401- 00200,
NAIROBI.

GEOTECHNICAL BOREHOLE LOG

ID: BH1 | **SHEET No. 1 of 3**

PROJECT : PROPOSED CONSTRUCTION OF KIRIAINI AHP

LOCATION : MATHIOYA CONSTITUENCY, MURANG'A COUNTY

CLIENT : STATE DEPARTMENT FOR HOUSING AND URBAN DEVELOPMENT

BOREHOLE DETAILS	
BOREHOLE DEPTH (m)	25.0
GROUND ELEVATION (m)	1883.4
COORDINATES (UTM) E	36°56'10.50"E
COORDINATES (UTM) N	0°36'32.90"S
COORDINATES (UTM) ZONE	37M
Boring Date(s) Start	4/JUNE/2026
Boring Date(s) End	5/JUNE/2026

PENETRATION METHOD: Rotary	Drilled By: Samuel Murimi	Bit Type:Diamond	
RIG TYPE: GY-150	Logged By: Josphat Nganga	Inclination From Vertical:	
AUGUR Ø (mm): 150 mm	Checked By: Anthony Murage	Final Borehole Water Level (m):	

Depth (m)	Elevation (masl)	Core Details: Run (m)	TCR (%)	RQD (%)	Lithology Log	Description	Weathering Grade	Fracture Index	N- VALUES
0	1883.4	1.0	100	NIL					9
1	1882.4	1.0	100	NIL					
2	1881.4	1.0	100	40					11
3	1880.4	1.0	100	NIL					
4	1879.4	1.0	100	NIL		Brown, medium grained GRAVELLY ELASTIC SILT	VI		14
5	1878.4	1.0	100	NIL					
6	1877.4	1.0	100	NIL					16
7	1876.4	1.0	100	NIL					
8	1875.4	1.0	100	NIL					19
9	1874.4	1.0	100	NIL					
10	1873.4								23
									25

SPT N-Value: 0 10 20 30 40 50

LEGEND

SILT SAND

WEATHERING GRADES

- I: Fresh & Hard
- II: Slightly Weathered
- III: Moderately Weathered
- IV: Highly Weathered
- V: Completely Weathered
- VI: Residual Soil

COHESIONLESS SOILS

Relative Density	N - Value
Very Loose	< 4
Loose	4 - 10
Medium Dense	10 - 30
Dense	30 - 50
Very Dense	> 50

COHESIVE SOILS

Consistency	N - Value
Very Soft	< 2
Soft	2 - 4
Firm	4 - 8
Stiff	8 - 16
Very Stiff	16 - 32
Hard	> 32

Cransfield Materials Testing Center

CRANSFIELD CENTER
P.O Box 59401- 00200,
NAIROBI.

GEOTECHNICAL BOREHOLE LOG

ID: BH1 | **SHEET No. 2 of 3**

PROJECT : PROPOSED CONSTRUCTION OF KIRIAINI AHP

LOCATION : MATHIOYA CONSTITUENCY, MURANG'A COUNTY

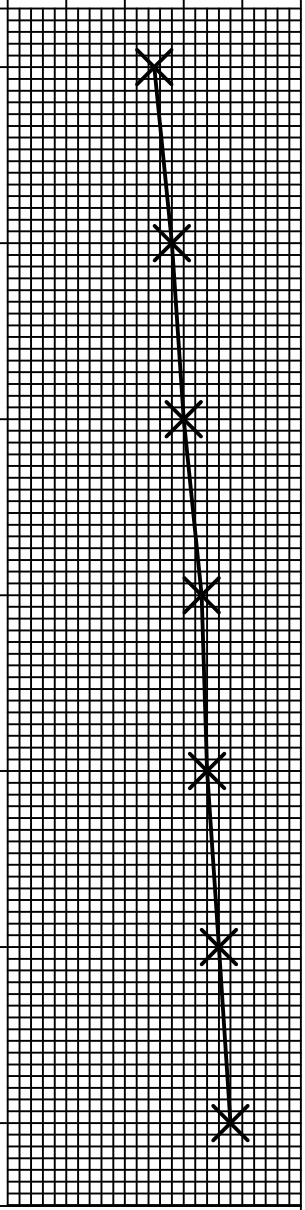
CLIENT : STATE DEPARTMENT FOR HOUSING AND URBAN DEVELOPMENT

BOREHOLE DETAILS	
BOREHOLE DEPTH (m)	25.0
GROUND ELEVATION (m)	1883.4
COORDINATES (UTM) E	36°56'10.50"E
COORDINATES (UTM) N	0°36'32.90"S
COORDINATES (UTM) ZONE	37M
Boring Date(s) Start	4/JUNE/2026
Boring Date(s) End	5/JUNE/2026

PENETRATION METHOD: Rotary	Drilled By: Samuel Murimi	Bit Type:Diamond	
RIG TYPE: GY-150	Logged By: Josphat Nganga	Inclination From Vertical:	
AUGUR Ø (mm): 150 mm	Checked By: Anthony Murage	Final Borehole Water Level (m):	

Depth (m)	Elevation (masl)	Core Details: Run (m)	TCR (%)	RQD (%)	Lithology Log	Description	Weathering Grade	Fracture Index	N- VALUES
10	1873.4	1.0	100	NIL					
11	1872.4	1.0	100	NIL					28
12	1871.4	1.0	100	40					30
13	1870.4	1.0	100	NIL					
14	1869.4	1.0	100	NIL					33
15	1868.4	1.0	100	NIL		Brown, medium grained GRAVELLY ELASTIC SILT	VI		
16	1867.4	1.0	100	NIL					34
17	1866.4	1.0	100	NIL					
18	1865.4	1.0	100	NIL					36
19	1864.4	1.0	100	NIL					
20	1863.4	1.0	100	NIL					38

SPT N-Value: 0 10 20 30 40 50



LEGEND

SILT SAND

WEATHERING GRADES

- I: Fresh & Hard
- II: Slightly Weathered
- III: Moderately Weathered
- IV: Highly Weathered
- V: Completely Weathered
- VI: Residual Soil

COHESIONLESS SOILS

Relative Density	N - Value
Very Loose	< 4
Loose	4 - 10
Medium Dense	10 - 30
Dense	30 - 50
Very Dense	> 50

COHESIVE SOILS

Consistency	N - Value
Very Soft	< 2
Soft	2 - 4
Firm	4 - 8
Stiff	8 - 16
Very Stiff	16 - 32
Hard	> 32

Cransfield Materials Testing Center

CRANSFIELD CENTER
P.O Box 59401- 00200,
NAIROBI.

GEOTECHNICAL BOREHOLE LOG

ID: BH1 | SHEET No. 3 of 3

PROJECT : PROPOSED CONSTRUCTION OF KIRIAINI AHP

LOCATION : MATHIOYA CONSTITUENCY, MURANG'A COUNTY

CLIENT : STATE DEPARTMENT FOR HOUSING AND URBAN DEVELOPMENT

BOREHOLE DETAILS	
BOREHOLE DEPTH (m)	25.0
GROUND ELEVATION (m)	1883.4
COORDINATES (UTM) E	36°56'10.50"E
COORDINATES (UTM) N	0°36'32.90"S
COORDINATES (UTM) ZONE	37M
Boring Date(s) Start	4/JUNE/2026
Boring Date(s) End	5/JUNE/2026

PENETRATION METHOD:	Rotary	Drilled By:	Samuel Murimi	Bit Type:	Diamond
RIG TYPE:	GY-150	Logged By:	Josphat Nganga	Inclination From Vertical:	
AUGUR Ø (mm):	150 mm	Checked By:	Anthony Murage	Final Borehole Water Level (m):	

Depth (m)	Elevation (masl)	Core Details: Run (m)	TCR (%)	RQD (%)	Lithology Log	Description	Weathering Grade	Fracture Index	N- VALUES
20	1863.4	1.0	100	NIL					39
21	1862.4	1.0	100	NIL					40
22	1861.4	1.0	100	40					
23	1860.4	1.0	100	NIL		Brown, medium grained GRAVELLY ELASTIC SILT	VI		39
24	1859.4	1.0	100	NIL					
25	1858.4								
26									
27									
28									
29									
30									

SPT N-Value: 0 10 20 30 40 50

LEGEND

SILT SAND

WEATHERING GRADES

I: Fresh & Hard
II: Slightly Weathered
III: Moderately Weathered
IV: Highly Weathered
V: Completely Weathered
VI: Residual Soil

COHESIONLESS SOILS

Relative Density	N - Value
Very Loose	< 4
Loose	4 - 10
Medium Dense	10 - 30
Dense	30 - 50
Very Dense	> 50

COHESIVE SOILS

Consistency	N - Value
Very Soft	< 2
Soft	2 - 4
Firm	4 - 8
Stiff	8 - 16
Very Stiff	16 - 32
Hard	> 32

Cransfield Materials Testing Center

CRANSFIELD CENTER
P.O Box 59401- 00200,
NAIROBI.

GEOTECHNICAL BOREHOLE LOG

ID: BH2 | **SHEET No. 1 of 3**

PROJECT : PROPOSED CONSTRUCTION OF KIRIAINI AHP

LOCATION : MATHIOYA CONSTITUENCY, MURANG'A COUNTY

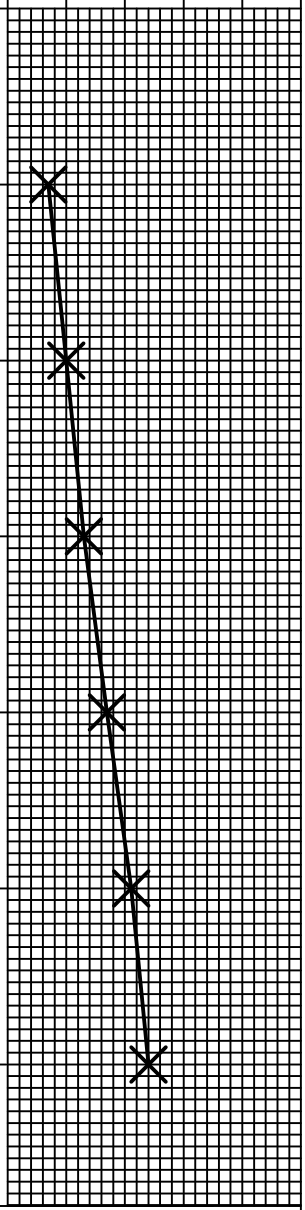
CLIENT : STATE DEPARTMENT FOR HOUSING AND URBAN DEVELOPMENT

BOREHOLE DETAILS	
BOREHOLE DEPTH (m)	25.0
GROUND ELEVATION (m)	1880.8
COORDINATES (UTM) E	36°56'10.70"E
COORDINATES (UTM) N	0°36'32.00"S
COORDINATES (UTM) ZONE	37M
Boring Date(s) Start	6/JUNE/2026
Boring Date(s) End	9/JUNE/2026

PENETRATION METHOD: Rotary	Drilled By: Samuel Murimi	Bit Type:Diamond	
RIG TYPE: GY-150	Logged By: Josphat Nganga	Inclination From Vertical:	
AUGUR Ø (mm): 150 mm	Checked By: Anthony Murage	Final Borehole Water Level (m):	

Depth (m)	Elevation (masl)	Core Details: Run (m)	TCR (%)	RQD (%)	Lithology Log	Description	Weathering Grade	Fracture Index	N- VALUES
0	1880.8								
		1.0	100	NIL					7
1	1879.8								
		1.0	100	NIL					
2	1878.8								10
		1.0	100	40					
3	1877.8								
		1.0	100	NIL					13
4	1876.8					Brown, medium grained GRAVELLY ELASTIC SILT WITH SAND	VI		
		1.0	100	NIL					
5	1875.8								17
		1.0	100	NIL					
6	1874.8								
		1.0	100	NIL					21
7	1873.8								
		1.0	100	NIL					
8	1872.8								24
		1.0	100	NIL					
9	1871.8								
		1.0	100	NIL					25
10	1870.8								

SPT N-Value: 0 10 20 30 40 50



LEGEND

SILT SAND

WEATHERING GRADES

- I: Fresh & Hard
- II: Slightly Weathered
- III: Moderately Weathered
- IV: Highly Weathered
- V: Completely Weathered
- VI: Residual Soil

COHESIONLESS SOILS

Relative Density	N - Value
Very Loose	< 4
Loose	4 - 10
Medium Dense	10 - 30
Dense	30 - 50
Very Dense	> 50

COHESIVE SOILS

Consistency	N - Value
Very Soft	< 2
Soft	2 - 4
Firm	4 - 8
Stiff	8 - 16
Very Stiff	16 - 32
Hard	> 32

Cransfield Materials Testing Center

CRANSFIELD CENTER
P.O Box 59401- 00200,
NAIROBI.

GEOTECHNICAL BOREHOLE LOG

ID: BH2 | **SHEET No. 2 of 3**

PROJECT : PROPOSED CONSTRUCTION OF KIRIAINI AHP

LOCATION : MATHIOYA CONSTITUENCY, MURANG'A COUNTY

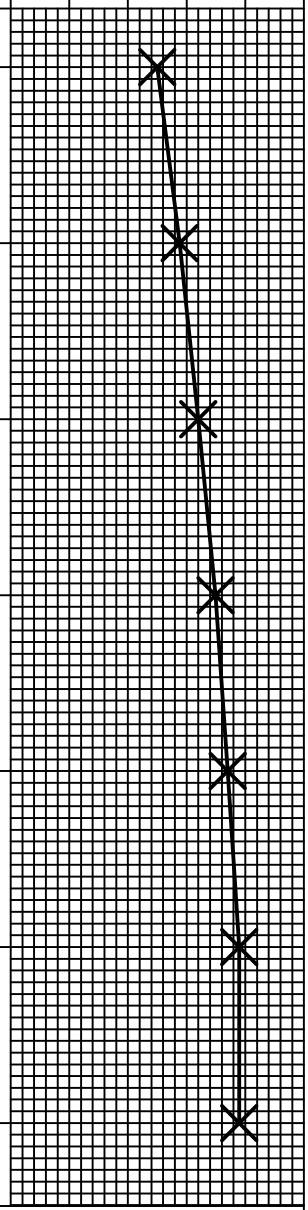
CLIENT : STATE DEPARTMENT FOR HOUSING AND URBAN DEVELOPMENT

BOREHOLE DETAILS	
BOREHOLE DEPTH (m)	25.0
GROUND ELEVATION (m)	1880.8
COORDINATES (UTM) E	36°56'10.70"E
COORDINATES (UTM) N	0°36'32.00"S
COORDINATES (UTM) ZONE	37M
Boring Date(s) Start	6/JUNE/2026
Boring Date(s) End	9/JUNE/2026

PENETRATION METHOD: Rotary	Drilled By: Samuel Murimi	Bit Type:Diamond	
RIG TYPE: GY-150	Logged By: Josphat Nganga	Inclination From Vertical:	
AUGUR Ø (mm): 150 mm	Checked By: Anthony Murage	Final Borehole Water Level (m):	

Depth (m)	Elevation (masl)	Core Details: Run (m)	TCR (%)	RQD (%)	Lithology Log	Description	Weathering Grade	Fracture Index	N- VALUES
10	1870.8	1.0	100	NIL		Brown, medium grained GRAVELLY ELASTIC SILT WITH SAND	VI		
11	1869.8	1.0	100	NIL					29
12	1868.8	1.0	100	40					32
13	1867.8	1.0	100	NIL					
14	1866.8	1.0	100	NIL					35
15	1865.8	1.0	100	NIL					37
16	1864.8	1.0	100	NIL					
17	1863.8	1.0	100	NIL					39
18	1862.8	1.0	100	NIL					
19	1861.8	1.0	100	NIL					39
20	1860.8								

SPT N-Value: 0 10 20 30 40 50



LEGEND

SILT SAND

WEATHERING GRADES

- I: Fresh & Hard
- II: Slightly Weathered
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- IV: Highly Weathered
- V: Completely Weathered
- VI: Residual Soil

COHESIONLESS SOILS

Relative Density	N - Value
Very Loose	< 4
Loose	4 - 10
Medium Dense	10 - 30
Dense	30 - 50
Very Dense	> 50

COHESIVE SOILS

Consistency	N - Value
Very Soft	< 2
Soft	2 - 4
Firm	4 - 8
Stiff	8 - 16
Very Stiff	16 - 32
Hard	> 32

Cransfield Materials Testing Center

CRANSFIELD CENTER
P.O Box 59401- 00200,
NAIROBI.

GEOTECHNICAL BOREHOLE LOG

ID: BH2 | **SHEET No. 3 of 3**

PROJECT : PROPOSED CONSTRUCTION OF KIRIAINI AHP

LOCATION : MATHIOYA CONSTITUENCY, MURANG'A COUNTY

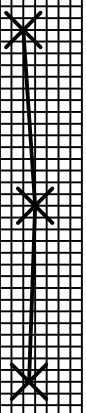
CLIENT : STATE DEPARTMENT FOR HOUSING AND URBAN DEVELOPMENT

BOREHOLE DETAILS	
BOREHOLE DEPTH (m)	25.0
GROUND ELEVATION (m)	1880.8
COORDINATES (UTM) E	36°56'10.70"E
COORDINATES (UTM) N	0°36'32.00"S
COORDINATES (UTM) ZONE	37M
Boring Date(s) Start	6/JUNE/2026
Boring Date(s) End	9/JUNE/2026

PENETRATION METHOD: Rotary	Drilled By: Samuel Murimi	Bit Type:Diamond	
RIG TYPE: GY-150	Logged By: Josphat Nganga	Inclination From Vertical:	
AUGUR Ø (mm): 150 mm	Checked By: Anthony Murage	Final Borehole Water Level (m):	

Depth (m)	Elevation (masl)	Core Details: Run (m)	TCR (%)	RQD (%)	Lithology Log	Description	Weathering Grade	Fracture Index	N- VALUES
20	1860.8	1.0	100	NIL					40
21	1861.8	1.0	100	NIL					42
22	1862.8	1.0	100	40		Brown, medium grained GRAVELLY ELASTIC SILT WITH SAND	VI		
23	1863.8	1.0	100	NIL					41
24	1864.8	1.0	100	NIL					
25	1865.8								
26									
27									
28									
29									
30									

SPT N-Value: 0 10 20 30 40 50



LEGEND

SILT SAND

WEATHERING GRADES

- I: Fresh & Hard
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Relative Density	N - Value
Very Loose	< 4
Loose	4 - 10
Medium Dense	10 - 30
Dense	30 - 50
Very Dense	> 50

COHESIVE SOILS

Consistency	N - Value
Very Soft	< 2
Soft	2 - 4
Firm	4 - 8
Stiff	8 - 16
Very Stiff	16 - 32
Hard	> 32

12. APPENDIX III (CORE BOX AND SITE PHOTOGRAPHS)

Borehole No 1 (1-10.0m)





Borehole No 1 (10.0-25.0m)





Borehole No 2 (1-15.0m)







Borehole No 2 (15.0-25.0m)





SITE PHOTOS





